

**UNITED STATES
SECURITIES AND EXCHANGE COMMISSION**

Washington, D.C. 20549

FORM 10-Q

(Mark One)

☒ **QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the quarterly period ended June 30, 2021

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from _____ to _____

Commission file number: 001-36481

ASPEN AEROGELS, INC.

(Exact name of registrant as specified in its charter)

Delaware

**(State or other jurisdiction of
incorporation or organization)**

**30 Forbes Road, Building B
Northborough, Massachusetts
(Address of principal executive offices)**

04-3559972

**(I.R.S. Employer
Identification No.)**

01532

(Zip Code)

Registrant's telephone number, including area code: (508) 691-1111

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of exchange on which registered
Common Stock, par value \$0.00001 per share	ASPN	The New York Stock Exchange

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days.

Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer

☐

Non-accelerated filer

☐

Accelerated filer

☒

Smaller reporting company

☒

Emerging growth company

☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

As of August 3, 2021, the registrant had 32,943,851 shares of common stock outstanding.

ASPEN AEROGELS, INC.
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Trademarks, Trade Names and Service Marks

We own or have rights to use "Aspen Aerogels," "Cryogel," "Pyrogel," "Spaceloft," "PyroThin," the Aspen Aerogels logo and other trademarks, service marks and trade names of Aspen Aerogels, Inc. appearing in this Quarterly Report on Form 10-Q. Solely for convenience, the trademarks, service marks and trade names referred to in this report are presented without the ® and TM symbols, but such references are not intended to indicate, in any way, that the owner thereof will not assert, to the fullest extent under applicable law, such owner's rights to these trademarks, service marks and trade names. This report contains additional trademarks, service marks and trade names of other companies, which, to our knowledge, are the property of their respective owners.

PART I — FINANCIAL INFORMATION

Item 1. Financial Statements.

ASPEN AEROGELS, INC.

Consolidated Balance Sheets

(Unaudited)

	June 30, 2021	December 31, 2020
	(In thousands, except share and per share data)	
Assets		
Current assets:		
Cash and cash equivalents	\$ 102,257	\$ 16,496
Accounts receivable, net of allowances of \$142 and \$442	19,366	15,698
Inventories	9,438	13,099
Prepaid expenses and other current assets	2,071	1,830
Total current assets	133,132	47,123
Property, plant and equipment, net	46,085	46,739
Operating lease right-of-use assets	11,990	3,478
Other long-term assets	1,017	84
Total assets	<u>\$ 192,224</u>	<u>\$ 97,424</u>
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	\$ 9,522	\$ 5,351
Accrued expenses	6,967	3,884
Current portion of long-term debt	3,675	1,609
Current portion of prepayment liability	4,530	—
Deferred revenue	1,870	2,037
Operating lease liabilities	1,596	1,046
Total current liabilities	28,160	13,927
Prepayment liability	5,000	9,555
Long-term debt	—	2,059
Operating lease liabilities long-term	11,482	3,597
Other long-term liabilities	434	434
Total liabilities	45,076	29,572
Commitments and contingencies (Note 9)		
Stockholders' equity:		
Preferred stock, \$0.00001 par value; 5,000,000 shares authorized, no shares issued and outstanding at June 30, 2021 and December 31, 2020	—	—
Common stock, \$0.00001 par value; 125,000,000 shares authorized, 32,907,101 and 27,821,685 shares issued and outstanding at June 30, 2021 and December 31, 2020, respectively	—	—
Additional paid-in capital	668,026	575,811
Accumulated deficit	(520,878)	(507,959)
Total stockholders' equity	147,148	67,852
Total liabilities and stockholders' equity	<u>\$ 192,224</u>	<u>\$ 97,424</u>

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Operations
(Unaudited)

	Three Months Ended June 30,		Six Months Ended June 30,	
	2021	2020	2021	2020
(In thousands, except share and per share data)				
Revenue:				
Product	\$ 31,490	\$ 24,526	\$ 59,546	\$ 52,833
Research services	180	115	221	227
Total revenue	31,670	24,641	59,767	53,060
Cost of revenue:				
Product	27,051	21,761	51,180	44,160
Research services	39	29	51	69
Gross profit	4,580	2,851	8,536	8,831
Operating expenses:				
Research and development	2,609	2,121	5,051	4,348
Sales and marketing	3,568	2,972	6,869	6,296
General and administrative	5,017	3,406	9,405	6,921
Total operating expenses	11,194	8,499	21,325	17,565
Loss from operations	(6,614)	(5,648)	(12,789)	(8,734)
Interest expense, net	(55)	(50)	(130)	(133)
Total interest expense, net	(55)	(50)	(130)	(133)
Net loss	<u>\$ (6,669)</u>	<u>\$ (5,698)</u>	<u>\$ (12,919)</u>	<u>\$ (8,867)</u>
Net loss per share:				
Basic and diluted	<u>\$ (0.23)</u>	<u>\$ (0.21)</u>	<u>\$ (0.46)</u>	<u>\$ (0.34)</u>
Weighted-average common shares outstanding:				
Basic and diluted	<u>28,501,044</u>	<u>26,521,861</u>	<u>28,243,687</u>	<u>25,858,076</u>

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.

Consolidated Statements of Stockholders' Equity

(Unaudited)

(In thousands, except share data)

	Preferred Stock \$0.00001 Par Value		Common Stock \$0.00001 Par Value		Additional Paid-in Capital	Accumulated Deficit	Total Stockholders' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2020	—	\$ —	27,821,685	\$ —	\$ 575,811	\$ (507,959)	\$ 67,852
Net loss	—	—	—	—	—	(6,250)	(6,250)
Stock compensation expense	—	—	—	—	976	—	976
Vesting of restricted stock units	—	—	246,737	—	(2,613)	—	(2,613)
Proceeds from employee stock option exercises	—	—	48,056	—	463	—	463
Proceeds from at-the-market offering, net of commissions and fees of \$193 and issuance costs of \$17	—	—	305,182	—	6,215	—	6,215
Forfeiture of performance-based restricted stock	—	—	(78,125)	—	—	—	—
Balance at March 31, 2021	—	\$ —	28,343,535	\$ —	\$ 580,852	\$ (514,209)	\$ 66,643
Net loss	—	—	—	—	—	(6,669)	(6,669)
Stock compensation expense	—	—	—	—	1,070	—	1,070
Issuance of restricted stock	—	—	476,550	—	—	—	—
Vesting of restricted stock units	—	—	6,207	—	(64)	—	(64)
Proceeds from employee stock option exercises	—	—	23,886	—	230	—	230
Proceeds from at-the-market offering, net of commissions and fees of \$383 and issuance costs of \$27	—	—	594,799	—	12,352	—	12,352
Proceeds from private placement common stock offering, net of fees and issuance costs of \$1,414	—	—	3,462,124	—	73,586	—	73,586
Balance at June 30, 2021	—	\$ —	32,907,101	\$ —	\$ 668,026	\$ (520,878)	\$ 147,148

	Preferred Stock \$0.00001 Par Value		Common Stock \$0.00001 Par Value		Additional Paid-in Capital	Accumulated Deficit	Total Stockholders' Equity
	Shares	Value	Shares	Value			
Balance at December 31, 2019	—	\$ —	24,302,504	\$ —	\$ 545,140	\$ (486,150)	\$ 58,990
Net loss	—	—	—	—	—	(3,169)	(3,169)
Stock compensation expense	—	—	—	—	992	—	992
Vesting of restricted stock units	—	—	336,951	—	(1,195)	—	(1,195)
Proceeds from underwritten public offering, net of underwriting discounts and commissions of \$1,093 and issuance costs of \$285	—	—	1,955,000	—	14,751	—	14,751
Balance at March 31, 2020	—	\$ —	26,594,455	\$ —	\$ 559,688	\$ (489,319)	\$ 70,369
Net loss	—	—	—	—	—	(5,698)	(5,698)
Stock compensation expense	—	—	—	—	1,007	—	1,007
Issuance of restricted stock	—	—	45,066	—	—	—	—
Vesting of restricted stock units	—	—	5,629	—	(16)	—	(16)
Proceeds from employee stock option exercises	—	—	200,159	—	869	—	869
Balance at June 30, 2020	—	\$ —	26,845,309	\$ —	\$ 561,548	\$ (495,017)	\$ 66,531

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
Consolidated Statements of Cash Flows
(Unaudited)

	Six Months Ended June 30,	
	2021	2020
	(In thousands)	
Cash flows from operating activities:		
Net loss	\$ (12,919)	\$ (8,867)
Adjustments to reconcile net loss to net cash used in operating activities:		
Depreciation	4,742	5,125
Amortization of debt issuance costs	7	2
Provision for bad debt	(98)	—
Stock-compensation expense	2,046	1,999
Reduction in the carrying amount of operating lease right-of-use assets	520	472
Changes in operating assets and liabilities:		
Accounts receivable	(3,570)	13,086
Inventories	3,661	(684)
Prepaid expenses and other assets	(1,174)	(287)
Accounts payable	3,962	(7,546)
Accrued expenses	3,083	(3,965)
Deferred revenue	(192)	(2,318)
Operating lease liabilities	(597)	(537)
Other liabilities	—	278
Net cash used in operating activities	(529)	(3,242)
Cash flows from investing activities:		
Capital expenditures	(3,879)	(1,977)
Net cash used in investing activities	(3,879)	(1,977)
Cash flows from financing activities:		
Proceeds from underwritten public offering, net of underwriting discounts and commissions of \$1,093	—	15,036
Issuance costs from underwritten public offering	—	(285)
Proceeds from issuance of long-term debt	—	3,686
Issuance costs from long-term debt	—	(27)
Repayments of borrowings under line of credit, net	—	(3,123)
Proceeds from employee stock option exercises	693	869
Payments made for employee restricted stock tax withholdings	(2,677)	(1,211)
Proceeds from at-the-market offering, net of commissions and fees of \$576	18,611	—
Issuance costs from at-the-market offering	(44)	—
Proceeds from private placement offering	75,000	—
Fees and issuance costs from private placement offering	(1,414)	—
Net cash provided by financing activities	90,169	14,945
Net increase in cash	85,761	9,726
Cash and cash equivalents at beginning of period	16,496	3,633
Cash and cash equivalents at end of period	\$ 102,257	\$ 13,359
Supplemental disclosures of cash flow information:		
Interest paid	\$ 113	\$ 125
Income taxes paid	\$ —	\$ —
Supplemental disclosures of non-cash activities:		
Right-of-use assets obtained in exchange for new operating lease liabilities	\$ 9,032	\$ 389
Changes in accrued capital expenditures	\$ 209	\$ (337)

See accompanying notes to unaudited consolidated financial statements.

ASPEN AEROGELS, INC.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(UNAUDITED)

(1) Description of Business and Basis of Presentation

Nature of Business

Aspen Aerogels, Inc. (the Company) is an aerogel technology company that designs, develops and manufactures innovative, high-performance aerogel insulation used primarily in the energy infrastructure and building materials markets. In addition, the Company has introduced a line of aerogel thermal barriers for use in lithium-ion batteries in the electric vehicle market. The Company is also developing applications for its aerogel technology in the battery materials and a number of other high-potential markets.

The Company also conducts research related to aerogel technology supported by funding from several agencies of the U.S. government and other institutions in the form of research contracts. The Company has decided to cease efforts to secure additional funded research contracts and to wind down existing contract research activities.

The Company maintains its corporate offices in Northborough, Massachusetts. The Company has three wholly owned subsidiaries: Aspen Aerogels Rhode Island, LLC, Aspen Aerogels Germany, GmbH and Aspen Aerogels Georgia, LLC.

Liquidity

During the six months ended June 30, 2021, the Company incurred a net loss of \$12.9 million, used \$0.5 million of cash in operations, used \$3.9 million of cash for capital expenditures, received net proceeds of \$18.6 million through an at-the-market offering of the Company's common stock and received net proceeds of \$73.6 million through a private placement offering of the Company's common stock. At June 30, 2021, the Company had cash and cash equivalents of \$102.3 million, total debt of \$3.7 million, a \$4.5 million current prepayment liability (see note 9) and no outstanding borrowings under its revolving line of credit (see note 7). After giving effect to \$1.5 million of outstanding letters of credit, the amount available to the Company at June 30, 2021 under the revolving line of credit was \$7.9 million. The existing revolving line of credit matures on April 28, 2022.

The Company is increasing investment in the research and development of next-generation aerogel products and manufacturing process technologies. In addition, the Company is continuing to develop aerogel products and technologies for the electric vehicle market. The Company believes that the commercial potential for the Company's technology in the electric vehicle market is significant. Accordingly, the Company plans to continue to hire additional personnel, incur additional operating expenses, and incur capital expenditures to expand silica aerogel manufacturing capacity, build an automated thermal barrier fabrication operation, enhance research and development laboratory facilities and equipment, and construct a battery materials facility, among other efforts.

The Company expects its existing cash balance and the amount anticipated to be available under the existing revolving line of credit will be sufficient to support current operating requirements, current research and development activities and the initial capital expenditures required to support the evolving commercial opportunity in the electric vehicle market and other strategic business opportunities. However, the Company plans to supplement its cash balance and available credit with equity financings, debt financings, customer prepayments, or technology licensing fees to provide the additional capital necessary to complete future capacity expansions or to fund evolving strategic business initiatives.

Unaudited Interim Financial Information

The accompanying unaudited interim consolidated financial statements include the accounts of the Company and have been prepared in accordance with accounting principles generally accepted in the United States of America (U.S. GAAP) and with the instructions to Form 10-Q and Article 10 of Regulation S-X. Certain information and disclosures normally included in the consolidated financial statements prepared in accordance with U.S. GAAP have been condensed or omitted pursuant to such rules and regulations. As such, the information included in this Quarterly Report on Form 10-Q should be read in conjunction with the audited consolidated financial statements and accompanying notes in our Annual Report on Form 10-K for the year ended December 31, 2020 (the Annual Report), filed with the U.S. Securities and Exchange Commission on March 12, 2021.

In the opinion of the Company's management, the unaudited interim consolidated financial statements have been prepared on the same basis as the audited consolidated financial statements and include all adjustments that are of a normal recurring nature and necessary for the fair statement of the Company's financial position as of June 30, 2021 and the results of its operations and stockholders' equity for the three and six months ended June 30, 2021 and 2020 and the cash flows for the six month periods then ended. The Company has evaluated subsequent events through the date of this filing.

The Company's results of operations for the three and six months ended June 30, 2021 are not necessarily indicative of the results to be expected for the year ending December 31, 2021 or any other period. In addition, the Company is uncertain of the continued duration and severity of the COVID-19 pandemic and the impact it will have on the Company's results of operations for the year ending December 31, 2021 or any other period.

(2) Significant Accounting Policies

Principles of Consolidation

The accompanying consolidated financial statements, which have been prepared in accordance with U.S. GAAP, include the accounts of the Company and its wholly owned subsidiaries. All intercompany balances and transactions have been eliminated in consolidation.

Use of Estimates

The preparation of the consolidated financial statements requires the Company to make a number of estimates and assumptions that affect the reported amounts of assets and liabilities at the date of the consolidated financial statements and the reported amounts of revenues and expenses during the reporting period. Significant items subject to such estimates and assumptions include allowances for doubtful accounts, sales returns and allowances, product warranty costs, inventory valuation, the carrying amount of property and equipment, stock-based compensation and deferred income taxes. The Company evaluates its estimates and assumptions on an on-going basis using historical experience and other factors, including current economic conditions, which are believed to be reasonable under the circumstances. Management adjusts such estimates and assumptions when facts and circumstances warrant. Illiquid credit markets, volatile equity markets and declines in business investment can increase the uncertainty inherent in such estimates and assumptions. As future events and their effects cannot be determined with precision, actual results could differ significantly from these estimates. Changes in these estimates resulting from continuing changes in the economic environment will be reflected in the financial statements in future periods.

Cash and Cash Equivalents

Cash equivalents include short-term, highly liquid instruments, which consist of money market accounts. All cash and cash equivalents are maintained with major financial institutions in North America. Deposits with these financial institutions may exceed the amount of insurance provided on such deposits; however, these deposits typically may be redeemed upon demand and, therefore, bear minimal risk.

Concentration of Credit Risk

Financial instruments, which potentially expose the Company to concentrations of credit risk, consist principally of accounts receivable. The Company maintains an allowance for doubtful accounts based on its assessment of the collectability of accounts receivable. The Company reviews the allowance for doubtful accounts quarterly. During the six months ended June 30, 2021, the Company recorded a reduction for estimated customer uncollectible accounts receivable of \$0.1 million and had collections of \$0.2 million of previously reserved customer accounts receivables. The Company did not record a charge for uncollectible accounts receivable during the six months ended June 30, 2020.

Revenue Recognition

The Company recognizes revenue in accordance with Accounting Standards Codification 606, Revenue from Contracts with Customers (ASC 606). See note 3 for further details.

Leases

The Company accounts for its leases in accordance with Accounting Standards Update (ASU) 2016-02 (Topic 842). See note 10 for further details.

Stock-based Compensation

Stock-based compensation expense is measured at the grant date based on the fair value of the award. Expense is recognized on a straight-line basis over the requisite service period for all awards with service conditions. For performance-based awards, the grant date fair value is recognized as expense when the condition is probable of being achieved and then on a graded basis over the requisite service period. The Company uses the Black-Scholes option-pricing model to determine the fair value of service-based option awards, which requires a number of complex and subjective assumptions including the fair value of the underlying security, the expected volatility of the underlying security, a risk-free interest rate and the expected term of the option. The fair value of restricted stock and restricted stock unit grants is determined using the closing trading price of the Company's common stock on the date of grant. The fair value of awards containing market conditions is determined using a Monte-Carlo simulation model based upon the nature of the conditions, the expected volatility of the underlying security, and other relevant factors.

During the six months ended June 30, 2021, the Company granted 62,956 restricted common stock units (RSUs) with a grant date fair value of \$1.5 million and non-qualified stock options (NSOs) to purchase 202,189 shares of common stock with a grant date fair value of \$2.7 million to employees under the 2014 Employee, Director, and Consultant Equity Incentive Plan (the 2014 Equity Plan). The RSUs and NSOs granted to employees will vest over a three-year period. During the six months ended June 30, 2021, the Company also granted 14,934 shares of restricted common stock with a grant date fair value of \$0.3 million and NSOs to purchase 18,528 shares of common stock with a grant date fair value of \$0.2 million to its non-employee directors under the 2014 Equity Plan. The restricted common stock and NSOs granted to non-employee directors vest upon the earlier of the date that is the one-year anniversary of the grant or the day prior to the Company's annual meeting of stockholders to be held in 2022.

On June 29, 2021, the Company also issued 461,616 shares of restricted common stock to its Chief Executive Officer. The restricted common stock issued to the CEO will vest subject to achievement of certain common stock price targets, as defined, over a three-to-five year period. The Company used a Monte-Carlo simulation model to estimate the grant date fair value of the award. The equity award had an aggregate fair value of \$6.5 million at the time of grant.

Stock-based compensation is included in cost of revenue or operating expenses, as applicable, and consists of the following:

	Three Months Ended June 30,		Six Months Ended June 30,	
	2021	2020	2021	2020
(In thousands)				
Cost of product revenue	\$ 129	\$ 127	\$ 241	\$ 446
Research and development expenses	189	167	378	313
Sales and marketing expenses	206	174	374	345
General and administrative expenses	546	539	1,053	895
Total stock-based compensation	<u>\$ 1,070</u>	<u>\$ 1,007</u>	<u>\$ 2,046</u>	<u>\$ 1,999</u>

Pursuant to the "evergreen" provisions of the 2014 Equity Plan, the number of shares of common stock authorized for issuance under the plan automatically increased by 556,433 shares to 8,531,413 shares effective January 1, 2021.

As of June 30, 2021, 4,101,681 shares of common stock were reserved for issuance upon the exercise or vesting of outstanding stock-based awards granted under the 2014 Equity Plan. In addition, as of June 30, 2021, 79,960 shares of common stock were reserved for issuance upon the exercise of outstanding stock options granted under the Company's 2001 Equity Incentive Plan, as amended (the 2001 Equity Plan). Any cancellations or forfeitures of the options outstanding under the 2001 Equity Plan will result in the shares reserved for issuance upon exercise of such options becoming available for grant under the 2014 Equity Plan. As of June 30, 2021, the Company has either reserved in connection with statutory tax withholdings or issued a total of 3,632,924 shares under the 2014 Equity Plan. As of June 30, 2021, there were 716,848 shares of common stock available for future grant under the 2014 Equity Plan.

Net Loss per Share

The Company calculates net loss per share of common stock based on the weighted-average number of shares of common stock outstanding during each period. Potential common stock equivalents are determined using the treasury stock method. The weighted-average number of shares of common stock included in the computation of diluted net loss gives effect to all potentially dilutive common equivalent shares, including outstanding stock options and RSUs. Common equivalent shares are excluded from the computation of diluted net loss per share if their effect is antidilutive.

Segments

Operating segments are identified as components of an enterprise about which separate, discrete financial information is available for evaluation by the chief operating decision maker in making decisions on how to allocate resources and assess performance. The Company's chief operating decision maker is the Chief Executive Officer. The Company's chief operating decision maker reviews consolidated operating results to make decisions about allocating resources and assessing performance for the entire Company. The Company presently views its operations and manages its business as one operating segment.

Information about the Company's total revenues, based on shipment destination or services location, is presented in the following table:

	Three Months Ended		Six Months Ended	
	June 30,		June 30,	
	2021	2020	2021	2020
	(In thousands)			
Revenue:				
U.S.	\$ 15,544	\$ 8,092	\$ 28,299	\$ 21,765
International	16,126	16,549	31,468	31,295
Total	<u>\$ 31,670</u>	<u>\$ 24,641</u>	<u>\$ 59,767</u>	<u>\$ 53,060</u>

Warranty

The Company provides warranties for its products and records the estimated cost within cost of revenue in the period that the related revenue is recorded. The Company's standard warranty period extends to one year from the date of shipment. This standard warranty provides that the Company's products will be free from defects in material and workmanship, and will, under normal use, conform to the specifications for the product.

The Company's products may be utilized in systems that involve new technical demands and new configurations. Accordingly, the Company regularly reviews and assesses whether warranty reserves should be recorded in the period the related revenue is recorded.

The Company did not record any warranty expense during the six months ended June 30, 2021 and 2020.

Recently Issued Accounting Standards

From time to time, new accounting pronouncements are issued by the Financial Accounting Standards Board (FASB) or other standard setting bodies. Recently issued standards typically do not require adoption until a future effective date. Prior to their effective date, the Company evaluates the pronouncements to determine the potential effects of adoption to its consolidated financial statements.

Standards Implemented Since December 31, 2020

The Company has not implemented any accounting standards that had a material impact on its consolidated financial statements during the six months ended June 30, 2021.

Standards to be Implemented

The Company believes that the impact of recently issued accounting standards that are not yet effective will not have a material impact on its consolidated financial statements.

(3) Revenue from Contracts with Customers

Revenue Recognition

Revenue is recognized when a customer obtains control of promised goods or services in an amount that reflects the consideration which the entity expects to receive in exchange for those goods or services. To determine revenue recognition for arrangements within the scope of ASC 606, the Company performs the following five steps: (i) identification of the contract with a customer; (ii) identification of the performance obligations in the contract; (iii) determination of the transaction price; (iv) allocation of the transaction price to the separate performance obligations in the contract; and (v) recognition of the revenue associated with performance obligations as they are satisfied. The Company applies the five-step model to contracts when it is probable that the Company will collect the consideration it is entitled to in exchange for the goods or services it transfers to the customer. At contract inception, once the contract is determined to be within the scope of ASC 606, the Company assesses the goods or services promised within each contract and determines those that are performance obligations, and assesses whether each promised good or service is distinct. If the contract contains a single performance obligation, the entire transaction price is allocated to the single performance obligation. Contracts that contain multiple performance obligations require an allocation of the transaction price based on the estimated relative standalone-selling prices of the promised products or services underlying each performance obligation. The Company determines standalone-selling prices based on the price at which the performance obligation is sold separately. If the standalone-selling price is not observable through past transactions, the Company estimates the standalone-selling price taking into account available information such as market conditions and internally approved pricing guidelines related to the performance obligations. The Company then recognizes as revenue the amount of the transaction price that is allocated to the respective performance obligation when (or as) the performance obligation is satisfied.

When determining the transaction price of a contract, an adjustment is made if payment from a customer occurs either significantly before or significantly after performance, resulting in a significant financing component. Applying the practical expedient in paragraph ASC 606-10-32-18, the Company does not assess whether a significant financing component exists if the period between when the Company performs its obligations under the contract and when the customer pays is one year or less. The Company did not have any contracts outstanding at December 31, 2020 and did not enter into any contracts during the six months ended June 30, 2021 that contained a significant financing component.

The Company records deferred revenue for product sales when (i) the Company has delivered products but other revenue recognition criteria have not been satisfied or (ii) payments have been received in advance of the completion of required performance obligations.

Shipping and Handling Costs

Shipping and handling costs associated with outbound freight after control over a product has transferred to a customer are accounted for as fulfillment costs and are included in the cost of product revenue. The associated amount of revenue recognized includes the consideration to which the Company expects to be entitled to receive in exchange for incurring these shipping and handling costs.

Product Revenue

The Company generally enters into contracts containing one type of performance obligation. The Company recognizes product revenue when the performance obligation is satisfied, which is generally upon delivery according to contractual shipping terms within customer purchase orders.

The Company also enters into rebate agreements with certain customers. These agreements may be considered an additional performance obligation of the Company or variable consideration within a contract. Rebates are recorded as a reduction of revenue in the period the related product revenue is recognized. A corresponding liability is recorded as a component of deferred revenue on the consolidated balance sheets. These arrangements are primarily based on the customer attaining contractually specified sales volumes.

The Company estimates the amount of its product sales that may be returned by its customers and records this estimate as a reduction of revenue in the period the related product revenue is recognized. The Company currently estimates product return liabilities using historical rates of return, current quarter credit sales, and specific items of exposure on a contract-by-contract basis. Sales return reserves were approximately \$0.1 million at both June 30, 2021 and December 31, 2020.

Subsea Projects

The Company manufactures and sells subsea products that are designed for pipe-in-pipe applications in subsea oil production and are typically customized to meet customer specifications. Subsea products typically have no alternative use and contain an enforceable right to payment. Customer invoicing terms for subsea products are typically based on certain milestones within the production and delivery schedule. Under the provisions of ASC 606, the Company recognizes revenue at a point in time when transfer of control of the products is passed to the customer, or over time utilizing the input method. The timing of revenue recognition is assessed on a contract-by-contract basis. During the six months ended June 30, 2021 and 2020, the Company recognized revenue of \$1.1 million and \$5.3 million, respectively, in connection with subsea projects.

Research Services

The Company performs research services under contracts with various government agencies and other institutions. These contracts generally have one type of performance obligation associated with the provision of research services including certain licenses to any resulting intellectual property. The Company records revenue using the percentage-of-completion method in two ways: (1) for firm-fixed-price contracts, the Company accrues that portion of the total contract price that is allocable on the basis of the Company's estimates of costs incurred to date to total contract costs; and (2) for cost-plus-fixed-fee contracts, the Company records revenue that is equal to total payroll cost incurred times a stated factor plus reimbursable expenses, to a stated upper limit. The primary cost under the Company's research service contracts is the labor effort expended in completing the research. Typically, the only deliverable, other than the labor hours expended, is reporting research results to the customer or delivery of research grade aerogel products. Because the input measure of labor hours expended is also reflective of the output measure, it is a reliable means to measure the extent of progress toward completion. Revisions in cost estimates and fees during the course of the contract are reflected in the accounting period in which the facts that require the revisions become known. Contract costs and rates used to allocate overhead to contracts are subject to audit by the respective contracting government agency. Adjustments to revenue as a result of audit are recorded within the period they become known. To date, adjustments to revenue as a result of contracting agency audits have been insignificant.

Disaggregation of Revenue

In the following tables, revenue is disaggregated by primary geographical region and source of revenue:

	Three Months Ended June 30,					
	2021			2020		
	U.S.	International	Total	U.S.	International	Total
Geographical region	(In thousands)					
Asia	\$ —	\$ 4,553	\$ 4,553	\$ —	\$ 13,652	\$ 13,652
Canada	—	991	991	—	235	235
Europe	—	9,435	9,435	—	1,818	1,818
Latin America	—	1,147	1,147	—	844	844
U.S.	15,544	—	15,544	8,092	—	8,092
Total revenue	<u>\$ 15,544</u>	<u>\$ 16,126</u>	<u>\$ 31,670</u>	<u>\$ 8,092</u>	<u>\$ 16,549</u>	<u>\$ 24,641</u>
Source of revenue						
Product revenue	\$ 15,364	\$ 15,459	\$ 30,823	\$ 7,974	\$ 13,454	\$ 21,428
Subsea projects	—	667	667	3	3,095	3,098
Research services	180	—	180	115	—	115
Total revenue	<u>\$ 15,544</u>	<u>\$ 16,126</u>	<u>\$ 31,670</u>	<u>\$ 8,092</u>	<u>\$ 16,549</u>	<u>\$ 24,641</u>

	Six Months Ended June 30,					
	2021			2020		
	U.S.	International	Total	U.S.	International	Total
	(In thousands)					
Geographical region						
Asia	\$ —	\$ 10,141	\$ 10,141	\$ —	\$ 23,756	\$ 23,756
Canada	—	1,955	1,955	—	690	690
Europe	—	16,681	16,681	—	4,955	4,955
Latin America	—	2,691	2,691	—	1,894	1,894
U.S.	28,299	—	28,299	21,765	—	21,765
Total revenue	<u>\$ 28,299</u>	<u>\$ 31,468</u>	<u>\$ 59,767</u>	<u>\$ 21,765</u>	<u>\$ 31,295</u>	<u>\$ 53,060</u>
Source of revenue						
Product revenue	\$ 28,078	\$ 30,394	\$ 58,472	\$ 20,427	\$ 27,087	\$ 47,514
Subsea projects	—	1,074	1,074	1,111	4,208	5,319
Research services	221	—	221	227	—	227
Total revenue	<u>\$ 28,299</u>	<u>\$ 31,468</u>	<u>\$ 59,767</u>	<u>\$ 21,765</u>	<u>\$ 31,295</u>	<u>\$ 53,060</u>

Contract Balances

The following table presents changes in the Company's contract assets and contract liabilities during the six months ended June 30, 2021:

	Balance at December 31, 2020	Additions	Deductions	Balance at June 30, 2021
	(In thousands)			
Contract assets				
Subsea projects	\$ 1,370	\$ 1,707	\$ (1,770)	\$ 1,307
Research services	67	220	(179)	108
Total contract assets	<u>\$ 1,437</u>	<u>\$ 1,927</u>	<u>\$ (1,949)</u>	<u>\$ 1,415</u>
Contract liabilities				
Deferred revenue				
Product revenue	\$ 1,859	\$ 2,508	\$ (3,304)	\$ 1,063
Subsea projects	178	1,474	(845)	807
Prepayment liability	9,555	—	(25)	9,530
Total contract liabilities	<u>\$ 11,592</u>	<u>\$ 3,982</u>	<u>\$ (4,174)</u>	<u>\$ 11,400</u>

During the six months ended June 30, 2021, the Company recognized \$1.6 million of revenue that was included in deferred revenue at December 31, 2020.

A contract asset is recorded when the Company satisfies a performance obligation by transferring a promised good or service and has earned the right to consideration from its customer. These assets may represent a conditional or unconditional right to consideration and are included within accounts receivable on the consolidated balance sheets.

A contract liability is recorded when consideration is received, or such consideration is unconditionally due, from a customer prior to transferring goods or services under the terms of the contract. Contract liabilities are recognized as revenue after control of the products or services is transferred to the customer and all revenue recognition criteria have been met.

(4) Inventories

Inventories consist of the following:

	June 30, 2021	December 31, 2020
	(In thousands)	
Raw materials	\$ 5,013	\$ 4,068
Finished goods	4,425	9,031
Total	<u>\$ 9,438</u>	<u>\$ 13,099</u>

(5) Property, Plant and Equipment, Net

Property, plant and equipment consist of the following:

	June 30, 2021	December 31, 2020	Useful life
	(In thousands)		
Construction in progress	\$ 2,348	\$ 1,906	—
Buildings	24,016	24,016	30 years
Machinery and equipment	127,541	124,807	3-10 years
Computer equipment and software	9,303	8,850	3 years
Total	163,208	159,579	
Accumulated depreciation	(117,123)	(112,840)	
Property, plant and equipment, net	<u>\$ 46,085</u>	<u>\$ 46,739</u>	

Depreciation expense was \$4.7 million and \$5.1 million for the six months ended June 30, 2021 and 2020, respectively.

Construction in progress totaled \$2.3 million and \$1.9 million at June 30, 2021 and December 31, 2020, respectively, principally associated with capital projects in the Company's East Providence, Rhode Island facility.

(6) Accrued Expenses

Accrued expenses consist of the following:

	June 30, 2021	December 31, 2020
	(In thousands)	
Employee compensation	\$ 5,266	\$ 2,587
Other accrued expenses	1,701	1,297
Total	<u>\$ 6,967</u>	<u>\$ 3,884</u>

(7) Revolving Line of Credit

The Company is party to an Amended and Restated Loan and Security Agreement with Silicon Valley Bank (Loan Agreement). On March 12, 2021, the Loan Agreement was amended and restated to extend the maturity date of the revolving credit facility to April 28, 2022 and to establish certain minimum Adjusted EBITDA levels and minimum Adjusted Quick Ratio covenants, each as defined in the Loan Agreement.

Under the revolving credit facility, the Company is permitted to borrow a maximum of \$20.0 million, subject to continued covenant compliance and borrowing base requirements. The interest rate applicable to borrowings under the revolving credit facility is based on the prime rate, subject to a minimum rate of 4.00% per annum. The rates applicable to borrowings vary from prime rate plus 0.75% per annum to prime rate plus 2.00% per annum. In addition, the Company is required to pay a monthly unused revolving line facility fee of 0.50% per annum of the average unused portion of the revolving credit facility.

Under the Loan Agreement, the Company is required to comply with both non-financial and financial covenants, including a minimum Adjusted EBITDA covenant and a minimum Adjusted Quick Ratio covenant. At June 30, 2021, the Company was in compliance with all such covenants. Obligations under the Loan Agreement are secured by a security interest in all assets of the Company, including those at the East Providence facility, except for certain exclusions. The Company intends to extend or replace the facility prior to its maturity.

At June 30, 2021 and December 31, 2020, the Company had no amounts drawn from the revolving credit facility.

The Company has provided letters of credit to secure obligations under certain commercial contracts and other obligations. The Company had outstanding letters of credit backed by the revolving credit facility of \$1.5 million and \$1.4 million at June 30, 2021 and December 31, 2020, respectively, which reduce the funds otherwise available to the Company under the facility.

At June 30, 2021, the amount available to the Company under the revolving credit facility was \$7.9 million after giving effect to the \$1.5 million of outstanding letters of credit.

(8) Debt

On May 1, 2020, Aspen Aerogels Rhode Island, LLC (Borrower) executed a promissory note (Note) in favor of Northeast Bank to receive an unsecured loan in the principal amount of \$3.7 million (the PPP Loan) pursuant to the Paycheck Protection Program (PPP) established by the CARES Act, as amended by the Paycheck Protection Program Flexibility Act (Flexibility Act), and administered by the Small Business Administration (SBA). The Borrower conferred with representatives of the SBA prior to finalizing the PPP Loan. The PPP Loan was subsequently sold by Northeast Bank to The Loan Source, Inc. (PPP Investor), a secondary market investor.

The PPP Loan carries an interest rate of 1% per year and matures two years from the date of the Note. The PPP Loan indebtedness may be forgiven in whole or in part upon application by the Borrower to the PPP Investor. The PPP Investor will determine to what extent the PPP Loan is eligible for forgiveness, subject to SBA guidelines and other regulations, based on the use of loan proceeds for payroll costs, payment of interest on covered mortgage obligations, rent and utility costs over either an eight-week or 24-week period, at the Borrower's option, following the Borrower's receipt of the loan proceeds. Upon the Borrower's application for forgiveness, the SBA will review the Borrower's eligibility, use of proceeds and other certifications in connection with the application for the PPP Loan. Upon such review, the SBA may approve or deny the Borrower's loan forgiveness application, in whole or part. As of June 30, 2021, the Borrower had not applied for forgiveness.

If the Borrower has not applied for forgiveness within ten months from the end of the 24-week period following receipt of the loan proceeds, the Borrower will be required make payments of principal and accrued interest in equal monthly installments over the remaining term of the loan. In addition, the Flexibility Act permits the Borrower and the PPP Investor to mutually agree to extend the term of the PPP Loan to five years from the date of the Note. The Borrower may repay the PPP Loan at any time without penalty.

While the Borrower is not required to apply for forgiveness of the PPP Loan, upon application for forgiveness, the Borrower may not receive forgiveness of the PPP Loan in whole or in part. In addition, the amount of potential loan forgiveness may be reduced if the Borrower failed to maintain employee and salary levels during the applicable eight-week or 24-week period following receipt of the loan proceeds. If the Borrower applies for forgiveness, and the PPP Loan is not forgiven in whole or in part, the Borrower will be required to make payments of principal and accrued interest in equal monthly installments over the remaining term of the loan for the post-forgiveness balance outstanding.

The Note contains customary events of default relating to, among other things, payment defaults, breaches of representations and warranties, and defaults under any loan or agreement with another debtor, including the Company's credit facility with SVB, to the extent the PPP Investor believes such default may materially affect the Borrower's ability to repay the PPP Loan. The occurrence of an event of default, if not cured, may result in the Borrower's repayment of the PPP Loan prior to maturity.

The Borrower has used the proceeds of the PPP Loan to support ongoing operations and to sustain staffing levels in its East Providence, Rhode Island manufacturing facility despite the unfavorable impact of the COVID-19 pandemic and volatile energy markets on its business.

Long-term debt consists of the following:

	June 30, 2021	December 31, 2020
	(In thousands)	
Long-term debt, principal	\$ 3,686	\$ 3,686
Current portion of long-term debt	(3,675)	(1,609)
Debt issuance costs, net of accumulated amortization	(11)	(18)
Long-term debt	<u>\$ —</u>	<u>\$ 2,059</u>

The schedule of required principal payments remaining on long-term debt outstanding as of June 30, 2021 is as follows:

Year	Principal Payments (In thousands)
2021 (excluding the six months ended June 30, 2021)	1,609
2022	2,077
Total principal payments	<u>\$ 3,686</u>

(9) Commitments and Contingencies

Cloud Computing Agreement

The Company is party to a cloud computing agreement that is a service contract for enterprise resource planning software. The agreement has a three-year term beginning on January 15, 2021. During the six months ended June 30, 2021, the Company capitalized \$0.3 million of costs related to implementation of the agreement that will begin to amortize during 2022. The capitalized implementation costs are included within other long-term assets on the consolidated balance sheets.

Thermal Barrier Contracts

The Company is party to three contracts with a major U.S. automotive original equipment manufacturer (OEM) to supply fabricated, multi-part thermal barriers (Barriers) for use in the battery system of its next-generation electric vehicles (Contracts). Pursuant to the Contracts, the Company is obligated to supply Barriers at fixed annual prices and at volumes to be specified by the OEM up to a daily maximum quantity through the respective terms of the agreements, one of which expires in 2026 and two of which expire in 2034. While the OEM has agreed to purchase its requirement for Barriers from the Company for locations to be designated from time to time by the OEM, it has no obligation to purchase any minimum quantity of Barriers under the Contracts. In addition, the OEM may terminate the Contracts at any time and for any or no reason. All other terms of the Contracts are generally consistent with the OEM's standard purchase terms, including quality and warranty provisions customary in automotive industry.

BASF Supply Agreement

The Company is party to a supply agreement, as amended, with BASF Polyurethanes GmbH (BASF) (the Supply Agreement) and a joint development agreement with BASF SE (the JDA). Pursuant to the Supply Agreement, the Company will sell exclusively to BASF certain of the Company's products at annual volumes to be specified by BASF, subject to certain volume limits. However, BASF has no obligation to purchase products under the Supply Agreement. The Supply Agreement will terminate on December 31, 2027 with respect to the Company's Spaceloft A2 product and December 31, 2028 with respect to a new product developed under the JDA. Upon the expiration of the Supply Agreement with respect to each product, the Company will be subject to a post-termination

supply commitment for an additional two years. The JDA is designed to facilitate collaboration by the parties on the development and commercialization of new products.

In addition, BASF, in its sole discretion, may make prepayments to the Company in the aggregate amount of up to \$22.0 million during the term of the Supply Agreement. These prepayment obligations are secured by a security interest in real estate, plant and equipment at the Company's Rhode Island facility and a license to certain intellectual property. BASF made a prepayment in the amount of \$5.0 million to the Company in 2018 (the 2018 Prepayment). As of January 1, 2019, 25.3% of any amounts that the Company invoices for Spaceloft A2 sold to BASF are credited against the outstanding balance of the 2018 prepayment. If any of the 2018 Prepayment remains uncredited as of December 31, 2021, BASF may require that the Company repay the uncredited amount to BASF beginning in 2022.

Pursuant to the first addendum to the Supply Agreement, on January 30, 2019, BASF made an additional prepayment in the amount of \$5.0 million to the Company (the 2019 Prepayment). As of January 1, 2020, 50.0% of any amounts that the Company invoices for the newly developed product sold to BASF are credited against the outstanding balance of the 2019 Prepayment. After December 31, 2022, BASF may require that the Company credit an additional 24.7% of any amounts invoiced by the Company for Spaceloft A2 product sold to BASF against the outstanding balance of the 2019 Prepayment, if any, or may require that the Company repay the uncredited amount of the 2019 Prepayment to BASF in full.

As of June 30, 2021, the Company had received \$10.0 million in prepayments from BASF and applied approximately \$0.3 million of credits against amounts invoiced.

The prepayment liability consists of the following:

	June 30, 2021	December 31, 2020
	(In thousands)	
Prepayment liability	\$ 9,733	\$ 9,845
Current portion of prepayment liability	(4,530)	—
Prepayment liability included within deferred revenue	(203)	(290)
Prepayment liability, long-term	<u>\$ 5,000</u>	<u>\$ 9,555</u>

The amounts and terms of additional prepayment installments, if any, are subject to negotiation between the Company and BASF.

Federal, State and Local Environmental Regulations

The Company is subject to federal, state and local laws and regulations relating to the environment. These laws generally provide for control of pollutants released into the environment and require responsible parties to undertake remediation. Penalties may be imposed for noncompliance.

Litigation

The Company is, from time to time, a party to litigation that arises in the normal course of its business operations. See Part II, Item 1 "Legal Proceedings" of this Quarterly Report on Form 10-Q for a description of certain of the Company's current legal proceedings. The Company is not presently a party to any litigation for which it believes a loss is probable requiring an amount to be accrued or a possible loss contingency requiring disclosure.

(10) Leases

The Company leases office, laboratory, warehouse and fabrication space in Northborough, Massachusetts, Marlborough, Massachusetts, Pawtucket, Rhode Island, and East Providence, Rhode Island under operating leases. Under these agreements, the Company is obligated to pay annual rent, real estate taxes, and certain other operating expenses. The Company also leases equipment under operating leases. The Company's operating leases expire at various dates through 2031.

The Company determines if an arrangement is a lease at inception. ROU assets represent the Company's right to use an underlying asset for the lease term and lease liabilities represent the Company's payment obligations under the lease. Operating lease

ROU assets and liabilities are recognized based on the present value of lease payments over the lease term. To measure its lease liabilities, the Company uses its incremental borrowing rate or the rate implicit in the lease, if available. The Company calculates its incremental borrowing rate using a synthetic credit rating analysis based on Moody's Building Materials Industry Rating Methodology. ROU assets also include any direct costs and prepaid lease payments but exclude any lease incentives received. The Company's lease terms may include options to extend or terminate the lease when it is reasonably certain that the Company will exercise that option. Lease expense for lease payments is recognized on a straight-line basis over the lease term.

The Company elected the short-term lease recognition exemption for all leases that qualify. For leases that qualify for this exemption, the Company does not recognize ROU assets or lease liabilities. For lease agreements with lease and non-lease components, the Company accounts for each component separately. However, in the case of equipment leases, the Company accounts for lease and non-lease components as a single component.

Maturities of operating lease liabilities at June 30, 2021 are as follows:

Year	Operating Leases (In thousands)
2021 (excluding the six months ended June 30, 2021)	1,078
2022	2,560
2023	2,528
2024	1,938
2025	1,702
Thereafter	6,803
Total lease payments	16,609
Less imputed interest	(3,531)
Total lease liabilities	<u>\$ 13,078</u>

The Company incurred operating lease costs of \$0.7 million during both the six months ended June 30, 2021 and 2020. Cash payments related to operating lease liabilities were \$0.8 million and \$0.7 million during the six months ended June 30, 2021 and 2020, respectively.

At June 30, 2021, the weighted average remaining lease term for operating leases was 7.9 years. At June 30, 2021, the weighted average discount rate for operating leases was 6.3%.

As of June 30, 2021, the Company has additional operating equipment leases that will commence during 2021 with total lease payments of \$0.4 million and a weighted average lease term of 5.0 years.

(11) CARES Act Payroll Tax Deferral

The Company elected to defer approximately \$0.9 million of its employer payroll tax obligation for the period from March 27, 2020 to December 31, 2020 pursuant to the provisions of the CARES Act. The Company is required to remit 50 percent of the deferred tax balance on or before December 31, 2021 and the remaining 50 percent on or before December 31, 2022.

Other long-term liabilities consist of the following:

	June 30, 2021	December 31, 2020
	(In thousands)	
Deferred employer payroll tax obligation	\$ 870	\$ 870
Current portion of deferred payroll tax obligation	(436)	(436)
Other long-term liabilities	<u>\$ 434</u>	<u>\$ 434</u>

(12) Net Loss Per Share

The computation of basic and diluted net loss per share consists of the following:

	Three Months Ended June 30,		Six Months Ended June 30,	
	2021	2020	2021	2020
(In thousands, except share and per share data)				
Numerator:				
Net loss	\$ (6,669)	\$ (5,698)	\$ (12,919)	\$ (8,867)
Denominator:				
Weighted average shares outstanding, basic and diluted	28,501,044	26,521,861	28,243,687	25,858,076
Net loss per share, basic and diluted	\$ (0.23)	\$ (0.21)	\$ (0.46)	\$ (0.34)

Potentially dilutive common shares that were excluded from the computation of diluted net loss per share because they were anti-dilutive consist of the following:

	Three and Six Months Ended June 30,	
	2021	2020
Common stock options	3,834,009	3,954,223
Restricted common stock units	347,633	715,482
Restricted common stock awards	476,550	123,191
Total	4,658,192	4,792,896

In the table above, anti-dilutive shares consist of those common stock equivalents that have (i) an exercise price above the average stock price for the period or (ii) related average unrecognized stock compensation expense sufficient to buy back the entire amount of shares. The Company excludes the shares issued in connection with restricted stock awards from the calculation of basic weighted average common shares outstanding until the restrictions lapse.

(13) Income Taxes

The Company incurred net operating losses and recorded a full valuation allowance against net deferred tax assets for all periods presented. Accordingly, the Company has not recorded a provision for federal or state income taxes.

(14) Subsequent Events

The Company has evaluated subsequent events through August 4, 2021, the date of issuance of the consolidated financial statements for the three and six months ended June 30, 2021.

Item 2. Management's Discussion and Analysis of Financial Condition and Results of Operations.

The following information should be read in conjunction with the unaudited financial information and the notes thereto included in this Quarterly Report on Form 10-Q and the audited financial information and the notes thereto included in the Annual Report on Form 10-K for the year ended December 31, 2020, filed with the U.S. Securities and Exchange Commission (SEC) on March 12, 2021, which we refer to as the Annual Report.

Certain matters discussed in this Quarterly Report on Form 10-Q may be deemed to be forward-looking statements that involve risks and uncertainties. We make such forward-looking statements pursuant to the safe harbor provisions of the Private Securities Litigation Reform Act of 1995 and other federal securities laws. In this Quarterly Report on Form 10-Q, words such as "may," "will," "anticipate," "estimate," "expects," "projects," "intends," "plans," "believes" and similar expressions (as well as other words or expressions referencing future events, conditions or circumstances) are intended to identify forward-looking statements.

Our actual results and the timing of certain events may differ materially from the results discussed, projected, anticipated, or indicated in any forward-looking statements. We caution you that forward-looking statements are not guarantees of future performance and that our actual results of operations, financial condition and liquidity, and the development of the industry in which we operate may differ materially from the forward-looking statements contained in this Quarterly Report on Form 10-Q. In addition, even if our results of operations, financial condition and liquidity, and the development of the industry in which we operate are consistent with the forward-looking statements contained in this Quarterly Report on Form 10-Q, they may not be predictive of results or developments in future periods.

The following information and any forward-looking statements should be considered in light of factors discussed elsewhere in this Quarterly Report on Form 10-Q and under "Risk Factors" in Item 1A of the Annual Report.

We caution readers not to place undue reliance on any forward-looking statements made by us, which speak only as of the date they are made. We disclaim any obligation, except as specifically required by law and the rules of the SEC, to publicly update or revise any such statements to reflect any change in our expectations or in events, conditions or circumstances on which any such statements may be based, or that may affect the likelihood that actual results will differ from those set forth in the forward-looking statements.

You should read the following discussion and analysis of financial condition and results of operations together with Part I Item 1 "Financial Statements," which includes our financial statements and related notes, elsewhere in this Quarterly Report on Form 10-Q.

Investors and others should note that we routinely use the Investors section of our website to announce material information to investors and the marketplace. While not all of the information that we post on the Investors section of our website is of a material nature, some information could be deemed to be material. Accordingly, we encourage investors, the media, and others interested in us to review the information that we share on the Investors section of our website, <https://www.aerogel.com>.

Overview

We design, develop and manufacture innovative, high-performance aerogel insulation used primarily in the energy infrastructure and building materials markets. We believe our aerogel blankets deliver the best thermal performance of any widely used insulation product available on the market today and provide a combination of performance attributes unmatched by traditional insulation materials. Our end-use customers select our products where thermal performance is critical and to save money, improve resource efficiency, enhance sustainability, preserve operating assets and protect workers.

Our insulation is used by oil producers and the owners and operators of refineries, petrochemical plants, liquefied natural gas facilities, power generating assets and other energy infrastructure. Our Pyrogel and Cryogel product lines have undergone rigorous technical validation by industry leading end-users and achieved significant market adoption.

We are also actively developing a number of promising aerogel products and technologies for the electric vehicle market. We have developed and are commercializing our proprietary line of PyroThin thermal barriers for use in lithium-ion batteries in electric vehicles. Our PyroThin product is an ultra-thin, lightweight and flexible thermal barrier designed with other functional layers to impede the propagation of thermal runaway across multiple lithium-ion battery system architectures. Our thermal barrier technology is designed to offer a unique combination of performance attributes that enable electric vehicle manufacturers to achieve critical safety goals without sacrificing driving range. In addition, we are seeking to leverage our patented carbon aerogel technology to develop industry-leading battery materials for lithium-ion battery systems. These battery materials have the potential to enable an increase in the drive range of electric vehicles.

The commercial potential for our PyroThin thermal barriers and our carbon aerogel battery materials in the electric vehicle market is significant. Accordingly, we are planning to hire additional personnel, incur additional operating expenses, and incur capital expenditures to expand manufacturing capacity, build an automated thermal barrier fabrication operation, enhance research and development resources and construct a battery materials facility, among other items.

We also derive product revenue from a number of other end markets, including the building materials market. Customers in these markets use our products for applications as diverse as wall systems, military and commercial aircraft, trains, buses, appliances, apparel, footwear and outdoor gear. As we continue to enhance our aerogel technology platform, we believe we will have additional opportunities to address high-value applications in the global insulation market, the electric vehicle market and in a number of new, high-value markets.

We generate product revenue through the sale of our line of aerogel blankets and thermal barriers. We market and sell our products primarily through a sales force based in North America, Europe and Asia. The efforts of our sales force are supported by a small number of sales consultants with extensive knowledge of a particular market or region. Our sales force is responsible for establishing and maintaining customer and partner relationships, delivering highly technical information and ensuring high-quality customer service.

Our salespeople work directly with end-use customers and engineering firms to promote the qualification, specification and acceptance of our products. We also rely on an existing and well-established channel of qualified insulation distributors and contractors in more than 50 countries around the world to ensure rapid delivery of our products and strong end-user support.

We also perform research services under contracts with various agencies of the U.S. government, including the Department of Defense and the Department of Energy, and other institutions. We have decided to cease efforts to secure additional funded research contracts and to wind down our existing contract research activities. This decision reflected our desire to focus our research and development resources on initiatives to improve the profitability of our existing business and on efforts to develop new products and next-generation technology with application in new, potentially high-value markets.

We manufacture our products using our proprietary technology at our facility in East Providence, Rhode Island. We have operated the East Providence facility since 2008 and have increased our annual capacity in phases through December 31, 2020 to approximately 55 million square feet of aerogel blankets. We are currently engaged in an initiative, which we refer to as EP20, designed to increase the capacity of the East Providence facility to approximately 60 million square feet of aerogel blankets by the end of 2021. In addition, we anticipate that we will need to construct a state-of-the-art thermal barrier fabrication operation, hire dedicated thermal barrier fabrication employees, and increase our aerogel blanket manufacturing capacity to keep pace with the significant potential demand for our PyroThin thermal barriers. Accordingly, we are planning a significant expansion of our aerogel capacity prior to the end of 2023. The expected elements of the expansion plan will include the size of the required capacity expansion, the selection of an optimal manufacturing site for the expansion, and a detailed timeline for the construction and operation of the facility.

We have entered into three contracts with a major U.S. automotive original equipment manufacturer to supply fabricated, multi-part thermal barriers for use in the battery system of its next-generation electric vehicles. Pursuant to the contracts, we are obligated to supply the barriers at fixed annual prices and at volumes to be specified by the customer up to a daily maximum quantity through the respective term of the agreements, one of which expires in 2026 and two of which expire in 2034. While the customer has agreed to purchase from us its requirement for the barriers at locations to be designated from time to time, it has no obligation to purchase any minimum quantity of barriers under the contracts. In addition, the customer may terminate the contracts any time and for any or no reason. All other terms of the contracts are generally consistent with the customer's standard purchase terms, including quality and warranty provisions customary in the automotive industry.

We are engaged in a strategic partnership with BASF to develop and commercialize products for the building materials and other markets. The strategic partnership includes a supply agreement governing the exclusive sale of specified products to BASF and a joint development agreement targeting innovative products and technologies. BASF has no obligation to purchase any products under the supply agreement. Pursuant to the supply agreement, BASF may, in its sole discretion, make prepayments to us in the aggregate amount of up to \$22.0 million during the term of the agreement. We may repay the prepayments to BASF at any time in whole or in part for any reason.

BASF made a prepayment to us of \$5.0 million during 2018. As of January 1, 2019, 25.3% of any amounts that we invoice for Spaceloft A2 sold to BASF will be credited against the outstanding balance of the 2018 prepayment. If any amount of the 2018 prepayment remains uncredited at December 31, 2021, BASF may require that we repay the uncredited amount following a six-week notice period. In January 2019, BASF made an additional prepayment to us of \$5.0 million. As of January 1, 2020, 50% of any amounts that we invoice for a newly developed product sold to BASF will be credited against the outstanding balance of the 2019 prepayment. After December 31, 2022, BASF may require that we credit 24.7% of any amounts we invoice for Spaceloft A2 sold to

BASF against the outstanding balance of the 2019 prepayment or may require that we repay the uncredited amount following a six-week notice period.

On March 12, 2021, we entered into an Amended and Restated Loan and Security Agreement (Loan Agreement) with Silicon Valley Bank to extend the maturity date of the revolving credit facility to April 28, 2022. Pursuant to the Loan Agreement, we are permitted to borrow a maximum of \$20.0 million, subject to continued covenant compliance and borrowing base requirements. The interest rate applicable to borrowings under the revolving credit facility is based on the prime rate, as defined, subject to a minimum rate of 4.00% per annum. The rates applicable to borrowings vary from prime rate plus 0.75% per annum to prime rate plus 2.00% per annum. In addition, we are required to pay a monthly unused revolving line facility fee of 0.50% per annum of the average unused portion of the revolving credit facility. The credit facility has also been amended to establish certain minimum Adjusted EBITDA and minimum Adjusted Quick Ratio covenants, each as defined in the Loan Agreement.

On May 1, 2020, our wholly owned subsidiary, Aspen Aerogels Rhode Island, LLC (Borrower), executed a note for an unsecured loan of \$3.7 million pursuant to the Paycheck Protection Program (PPP Loan) under the Coronavirus Aid, Relief, and Economic Security Act (CARES Act), as amended, and administered by the U.S. Small Business Administration (SBA). The Borrower conferred with representatives of the SBA prior to finalizing the PPP Loan. The loan is unsecured, contains customary events of default, carries an interest rate of 1% per year, and matures on May 1, 2022. The Borrower may repay the loan at any time without penalty. In addition, the Borrower is permitted at any time to submit an application to extend the maturity of loan to May 1, 2025.

The Borrower may also choose to apply to have the PPP Loan forgiven in whole or in part subject to SBA guidelines. The potential amount of forgiveness is based on the Borrower's use of loan proceeds for payroll costs, mortgage interest payments, rent and utility costs over either an eight-week or 24-week period following receipt of the loan proceeds. The SBA may deny the loan forgiveness application if the agency determines that the Borrower was ineligible for the PPP Loan or any forgiveness criteria. As of June 30, 2021, the Borrower had not applied for forgiveness.

Upon application, the Borrower may receive loan forgiveness in whole or in part. In addition, the amount of potential loan forgiveness will be reduced if the Borrower failed to maintain employee and salary levels during the applicable eight-week or 24-week period following receipt of the loan proceeds. If the Borrower applies for forgiveness, and the PPP Loan is not forgiven in whole or in part, the Borrower will be required to begin to make payments of the principal and accrued interest of the post-forgiveness balance outstanding in equal monthly installments over the remaining term of the loan. If the Borrower does not apply for forgiveness by August 19, 2021, the Borrower will be required to make payments of principal and accrued interest in equal monthly installments over the remaining term of the loan.

The Borrower used the proceeds of the PPP Loan to support ongoing operations and to sustain staffing levels in the East Providence, Rhode Island manufacturing facility despite the unfavorable impact the COVID-19 pandemic and volatile energy markets had on its business.

On February 3, 2021, we entered into a supply agreement (Supply Agreement) with Silbond Corporation (Silbond), for the purchase of certain silanes (Product). Pursuant to the Supply Agreement, we agreed to purchase and Silbond agreed to supply, all of our requirements for the Product through the term of the Supply Agreement, which term ends on September 30, 2023, unless either party terminates the agreement early pursuant to the terms of the Supply Agreement.

On June 29, 2021, we entered into a securities purchase agreement (Purchase Agreement) with an affiliate of Koch Strategic Platforms (Purchaser). Pursuant to the terms of the Purchase Agreement, we sold to the Purchaser an aggregate of 3,462,124 shares of our common stock at a purchase price equal to \$21.663 per share, for aggregate gross proceeds of approximately \$75.0 million (the Private Placement).

Our revenue for the six months ended June 30, 2021 was \$59.8 million, which represented an increase of \$6.7 million, or 13%, from \$53.1 million for the six months ended June 30, 2020. Net loss for the six months ended June 30, 2021 was \$12.9 million and net loss per share was \$0.46. Net loss for the six months ended June 30, 2020 was \$8.9 million and net loss per share was \$0.34.

At present, we are not certain of the extent of the impact that the COVID-19 pandemic will continue to have on our business. Our manufacturing facility remains operational and we have not encountered any significant disruption to our supply chain or our ability to deliver to our customers.

In response to the COVID-19 pandemic, we have implemented and are following safe practices recommended by public health authorities and other government entities. We continue to focus on the safety and health of our employees, customers and vendors. In addition, we have implemented various precautionary measures, including remote work arrangements, restricted business travel and

procedures for social distancing, face coverings and safe hygiene. We continue to monitor public health guidance as it evolves and plan to adapt our practices as appropriate. Refer to “Risk Factors” in Item 1A of the Annual Report for more information concerning risks to our business associated with COVID-19.

Key Metrics and Non-GAAP Financial Measures

We regularly review a number of metrics, including the following key metrics, to evaluate our business, measure our performance, identify trends affecting our business, formulate financial projections and make strategic decisions.

Square Foot Operating Metric

We price our product and measure our product shipments in square feet. We estimate our annual capacity was approximately 55 million square feet of aerogel blankets at June 30, 2021. We believe the square foot operating metric allows us and our investors to measure our manufacturing capacity and product shipments on a uniform and consistent basis. The following chart sets forth product shipments in square feet associated with recognized revenue, including revenue recognized over time utilizing the input method, for the periods presented:

	Three Months Ended June 30,		Six Months Ended June 30,	
	2021	2020	2021	2020
	(In thousands)			
Product shipments in square feet	9,870	7,317	18,514	15,482

Adjusted EBITDA

We use Adjusted EBITDA, a non-GAAP financial measure, as a means to assess our operating performance. We define Adjusted EBITDA as net income (loss) before interest expense, taxes, depreciation, amortization, stock-based compensation expense and other items, from time to time, which we do not believe are indicative of our core operating performance. Adjusted EBITDA is a supplemental measure of our performance that is not presented in accordance with U.S. GAAP. Adjusted EBITDA should not be considered as an alternative to net income (loss) or any other measure of financial performance calculated and presented in accordance with U.S. GAAP. In addition, our definition and presentation of Adjusted EBITDA may not be comparable to similarly titled measures presented by other companies.

We use Adjusted EBITDA:

- as a measure of operating performance because it does not include the impact of items that we do not consider indicative of our core operating performance;
- for planning purposes, including the preparation of our annual operating budget;
- to allocate resources to enhance the financial performance of our business; and
- as a performance measure used under our bonus plan.

We also believe that the presentation of Adjusted EBITDA provides useful information to investors with respect to our results of operations and in assessing the performance and value of our business. Various measures of EBITDA are widely used by investors to measure a company’s operating performance without regard to items that can vary substantially from company to company depending upon financing and accounting methods, book values of assets, capital structures and the methods by which assets were acquired.

Although measures similar to Adjusted EBITDA are frequently used by investors and securities analysts in their evaluation of companies, we understand that Adjusted EBITDA has limitations as an analytical tool, and you should not consider it in isolation or as a substitute for net income (loss), income (loss) from operations, net cash provided by (used in) operating activities or an analysis of our results of operations as reported under U.S. GAAP. Some of these limitations are:

- Adjusted EBITDA does not reflect our historical cash expenditures or future requirements for capital expenditures or other contractual commitments;
- Adjusted EBITDA does not reflect changes in, or cash requirements for, our working capital needs;
- Adjusted EBITDA does not reflect stock-based compensation expense;

- Adjusted EBITDA does not reflect our income tax expense or cash requirements to pay our income taxes;
- Adjusted EBITDA does not reflect our interest expense, or the cash requirements necessary to service interest or principal payments on our debt;
- although depreciation, amortization and impairment charges are non-cash charges, the assets being depreciated, amortized or impaired will often have to be replaced in the future, and Adjusted EBITDA does not reflect any cash requirements for these replacements; and
- other companies in our industry may calculate EBITDA or Adjusted EBITDA differently than we do, limiting their usefulness as a comparative measure.

Because of these limitations, our Adjusted EBITDA should not be considered as a measure of discretionary cash available to us to reinvest in the growth of our business or as a measure of cash available for us to meet our obligations.

To properly and prudently evaluate our business, we encourage you to review the U.S. GAAP financial statements included elsewhere in this Quarterly Report on Form 10-Q, and not to rely on any single financial measure to evaluate our business.

The following table presents a reconciliation of net loss, the most directly comparable U.S. GAAP measure, to Adjusted EBITDA for the periods presented:

	Three Months Ended June 30,		Six Months Ended June 30,	
	2021	2020	2021	2020
	(In thousands)			
Net loss	\$ (6,669)	\$ (5,698)	\$ (12,919)	\$ (8,867)
Depreciation and amortization	2,104	2,562	4,742	5,125
Stock-based compensation ⁽¹⁾	1,070	1,007	2,046	1,999
Interest expense	55	50	130	133
Adjusted EBITDA	<u>\$ (3,440)</u>	<u>\$ (2,079)</u>	<u>\$ (6,001)</u>	<u>\$ (1,610)</u>

(1) Represents non-cash stock-based compensation related to vesting and modifications of stock option grants, vesting of restricted stock units and vesting of restricted common stock.

Our financial performance, including such measures as net income (loss), earnings per share and Adjusted EBITDA, are affected by a number of factors including volume and mix of aerogel products sold, average selling prices, our material costs and manufacturing expenses, the costs associated with capacity expansions and start-up of additional production capacity, and the amount and timing of operating expenses. Accordingly, we expect that our net income (loss), earnings per share and Adjusted EBITDA will vary from period to period.

During 2021, we are projecting growth in total revenue principally due to an anticipated increase in maintenance-based demand in the petrochemical and refinery market and growth in the European building materials market. We have also announced a price increase to offset an increase in raw material costs during 2021 that will contribute to revenue growth.

However, we intend to increase our investment in the electric vehicle market and our aerogel technology platform in 2021. We will use this investment to accelerate thermal barrier business development, to establish industry-leading thermal barrier fabrication capability, to progress from the development phase to the commercialization phase of our silicon-rich carbon aerogel battery materials, and to identify additional high-value markets for our aerogel technology, among other items. As a result, we expect to experience a decrease in Adjusted EBITDA and an increase in net loss versus 2020.

Components of Our Results of Operations

Revenue

We recognize product revenue from the sale of our line of aerogel products and research services revenue from the provision of services under contracts with various agencies of the U.S. government and other institutions. Product and research services revenue is recognized upon the satisfaction of contractual performance obligations.

We record deferred revenue for product sales when (i) we have delivered products but other revenue recognition criteria have not been satisfied or (ii) payments have been received in advance of the completion of required performance obligations.

During 2021, we are projecting growth in total revenue principally due to an anticipated increase in maintenance-based demand in the petrochemical and refinery market and growth in the European building materials market. We have also announced a price increase to offset an increase in raw material costs during 2021 that will contribute to revenue growth.

Cost of Revenue

Cost of product revenue consists primarily of materials and manufacturing expense. Cost of product revenue is recorded when the related product revenue is recognized.

Material is our most significant component of cost of product revenue and includes fibrous batting, silica materials and additives. Material costs as a percentage of product revenue vary from product to product due to differences in average selling prices, material requirements, product thicknesses and manufacturing yields. In addition, we provide warranties for our products and record the estimated cost within cost of revenue in the period that the related revenue is recorded or when we become aware that a potential warranty claim is probable and can be reasonably estimated. As a result of these factors, material costs as a percentage of product revenue will vary from period to period due to changes in the mix of aerogel products sold, the costs of our raw materials or the estimated cost of warranties. We expect that material costs will increase in absolute dollars due to projected growth in product shipments, but decrease as a percentage of revenue during 2021 due to the impact of our 2021 price increase, a projected favorable product mix and the impact of our bill of material cost initiatives.

Manufacturing expense is also a significant component of cost of revenue. Manufacturing expense includes labor, utilities, maintenance expense, and depreciation on manufacturing assets. Manufacturing expense also includes stock-based compensation for manufacturing employees and shipping costs. We expect that manufacturing expense will increase in absolute dollars and as a percentage of revenue during 2021 principally due to our plan to hire additional personnel and incur additional manufacturing expenses to establish fabrication operations in support of projected growth in PyroThin thermal barrier demand.

In total, we expect that cost of product revenue will increase in absolute dollars during 2021 versus 2020, but may increase or decrease modestly as a percentage of revenue versus 2020 depending on the level of revenue achieved during 2021.

Cost of research services revenue consists of direct labor costs of research personnel engaged in the contract research, third-party consulting and subcontractor expense, and associated direct material costs. This cost of revenue also includes overhead expenses associated with project resources, development tools and supplies. Cost of research services revenue is recorded when the related research services revenue is recognized. In 2021, we expect cost of research services revenue will decline as we wind down our existing contract research activities.

Gross Profit

Our gross profit as a percentage of revenue is affected by a number of factors, including the volume of aerogel products produced and sold, the mix of aerogel products sold, average selling prices, our material and manufacturing costs, realized capacity utilization and the costs associated with expansions and start-up of production capacity. Accordingly, we expect our gross profit in absolute dollars and as a percentage of revenue to vary significantly from period to period.

During 2021, we project that gross profit will grow in absolute dollars versus 2020 due to projected growth in total revenue and a decrease in projected material costs as a percentage of revenue, offset, in part, by an increase in manufacturing expense. However, we expect that gross profit as a percentage of revenue may increase or decrease modestly versus 2020 depending on the level of revenue achieved during 2021.

Operating Expenses

Operating expenses consist of research and development, sales and marketing, and general and administrative expenses. Operating expenses include personnel costs, legal fees, professional fees, service fees, insurance premiums, travel expense, facilities related costs and other costs, expenses and fees. The largest component of our operating expenses is personnel costs, consisting of salaries, benefits, incentive compensation and stock-based compensation. In any particular period, the timing and extent of personnel additions or reductions, legal activities, including patent enforcement actions, marketing programs, research efforts and a range of similar activities or actions could materially affect our operating expenses, both in absolute dollars and as a percentage of revenue.

During 2021, we expect to hire additional personnel and incur additional operating expenses to support the anticipated multi-year growth in our PyroThin thermal barrier business. As a result, we expect that operating expenses will increase in both absolute dollars and as a percentage of revenue during the year. In the longer term, we expect that operating expenses will increase in absolute dollars, but decrease as a percentage of revenue.

Research and Development Expenses

Research and development expenses consist primarily of expenses for personnel engaged in the development of next-generation aerogel compositions, form factors and manufacturing technologies. These expenses also include testing services, prototype expenses, consulting services, trial formulations for new products, equipment depreciation, facilities costs and related overhead. We expense research and development costs as incurred. We expect to continue to devote substantial resources to the development of new aerogel technologies, including our carbon aerogel battery materials. We believe that these investments are necessary to maintain and improve our competitive position. We also expect to continue to invest in research and engineering personnel and the infrastructure required in support of their efforts.

While we expect our research and development expenses will increase in absolute dollars but decrease as a percentage of revenue in the longer term, in 2021 we expect these expenses will increase in both absolute dollars and as a percentage of revenue.

Sales and Marketing Expenses

Sales and marketing expenses consist primarily of personnel costs, incentive compensation, marketing programs, travel and related costs, consulting expenses and facilities related costs.

We expect that sales and marketing expenses will increase in absolute dollars during 2021 principally due to an increase in compensation associated with the addition of personnel in support of our PyroThin thermal barrier business. In the longer term, we expect that sales and marketing expenses will increase in absolute dollars, but decrease as a percentage of revenue.

General and Administrative Expenses

General and administrative expenses consist primarily of personnel costs, legal expenses, consulting and professional services, audit and tax consulting costs, and expenses for our executive, finance, legal, human resources and information technology organizations. General and administrative expenses have increased as we have incurred additional costs related to operating as a publicly-traded company, which include costs of compliance with securities, corporate governance and related laws and regulations, investor relations expenses, increased insurance premiums, including director and officer insurance, and increased audit and legal fees.

We expect our general and administrative expenses to increase as we add general and administrative personnel to support the anticipated growth of our business. We also expect that the patent enforcement actions, described in more detail under “Legal Proceedings” in Part II, Item 1 of this Quarterly Report on Form 10-Q, if protracted, could result in significant legal expense over the medium to long-term. While we expect that our general and administrative expenses will increase in absolute dollars but decrease as a percentage of revenue in the longer term, in 2021 we expect such expenses will increase in both absolute dollars and as a percentage of revenue.

Interest Expense, Net

Interest expense, net consists primarily of fees and interest expense related to our revolving credit facility.

Provision for Income Taxes

We have incurred net losses since inception and have not recorded benefit provisions for U.S. federal income taxes or state income taxes since the tax benefits of our net losses have been offset by valuation allowances due to the uncertainty associated with the utilization of net operating loss carryforwards.

Results of Operations

Three months ended June 30, 2021 compared to the three months ended June 30, 2020

The following tables set forth a comparison of the components of our results of operations for the periods presented:

Revenue

	Three Months Ended June 30,				Change	
	2021		2020			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
(\$ in thousands)						
Revenue:						
Product	\$ 31,490	99%	\$ 24,526	100%	\$ 6,964	28%
Research services	180	1%	115	0%	65	57%
Total revenue	<u>\$ 31,670</u>	100%	<u>\$ 24,641</u>	100%	<u>\$ 7,029</u>	29%

The following chart sets forth product shipments in square feet associated with recognized revenue, including revenue recognized over time utilizing the input method, for the periods presented:

	Three Months Ended June 30,		Change	
	2021	2020	Amount	Percentage
Product shipments in square feet (in thousands)	9,870	7,317	2,553	35%

Total revenue increased \$7.1 million, or 29%, to \$31.7 million for the three months ended June 30, 2021 from \$24.6 million in the comparable period in 2020. The increase in total revenue was principally the result of an increase in product revenue.

Product revenue increased by \$7.0 million, or 28%, to \$31.5 million for the three months ended June 30, 2021 from \$24.5 million in the comparable period in 2020. This increase was principally driven by growth in the petrochemical and refinery markets, particularly in the United States, and growth in the European building materials market, offset, in part, by a decrease in project-based demand in the subsea and LNG markets.

Product revenue for the three months ended June 30, 2021 included \$9.6 million to a North American distributor and \$3.7 million to a European LNG project contractor. Product revenue for the three months ended June 30, 2020 included \$5.5 million to an Asian LNG project contractor, \$3.1 million to a subsea contractor and \$2.4 million to a North American distributor.

The average selling price per square foot of our products decreased by \$0.16, or 5%, to \$3.19 per square foot for the three months ended June 30, 2021 from \$3.35 per square foot for the three months ended June 30, 2020. The decrease in average selling price principally reflected the impact of a change in the mix of products sold. This decrease in average selling price had the effect of decreasing product revenue by \$1.6 million for the three months ended June 30, 2021 from the comparable period in 2020.

In volume terms, product shipments increased by 2.6 million square feet, or 35%, to 9.9 million square feet of aerogel products for the three months ended June 30, 2021, as compared to 7.3 million square feet for the three months ended June 30, 2020. The increase in product volume had the effect of increasing product revenue by \$8.6 million for the three months ended June 30, 2021 from the comparable period in 2020.

Research services revenue increased \$0.1 million, or 57%, to \$0.2 million for the three months ended June 30, 2021 from \$0.1 million in the comparable period in 2020. The increase was due to the timing of research work performed.

Product revenue was 99% of total revenue for the three months ended June 30, 2021 and greater than 99% of total revenue for the three months ended June 30, 2020. Research services revenue was 1% of total revenue for the three months ended June 30, 2021 and less than 1% of total revenue for the three months ended June 30, 2020.

Cost of Revenue

	Three Months Ended June 30,						Change	
	2021			2020				
	Amount	Percentage of Related Revenue	Percentage of Total Revenue	Amount	Percentage of Related Revenue	Percentage of Total Revenue	Amount	Percentage
(\$ in thousands)								
Cost of revenue:								
Product	\$ 27,051	86%	85%	\$ 21,761	89%	88%	\$ 5,290	24%
Research services	39	22%	0%	29	25%	0%	10	34%
Total cost of revenue	\$ 27,090	86%	86%	\$ 21,790	88%	88%	\$ 5,300	24%

Total cost of revenue increased \$5.3 million, or 24%, to \$27.1 million for the three months ended June 30, 2021 from \$21.8 million in the comparable period in 2020. The increase in total cost of revenue was primarily the result of an increase in product cost of revenue.

Product cost of revenue increased by \$5.3 million, or 24%, to \$27.1 million for the three months ended June 30, 2021 from \$21.8 million in the comparable period in 2020. The \$5.3 million increase was the result of a \$2.5 million increase in material costs and a \$2.8 million increase in manufacturing expense. The increase in material costs was principally the result of the 2.6 million square feet, or 35%, increase in total product shipments. The increase in manufacturing expense was the result of increases in compensation and related expenses of \$1.8 million, operating supplies of \$0.4 million, maintenance costs of \$0.3 million, waste disposal costs of \$0.2 million, utilities cost of \$0.2 million and professional services of \$0.1 million, offset, in part, by a decrease in other manufacturing expenses of \$0.2 million.

Product cost of revenue as a percentage of product revenue decreased to 86% during the three months ended June 30, 2021 from 89% during the three months ended June 30, 2020. This decrease was principally the result of the decrease in material costs as a percentage of revenue during the three months ended June 30, 2021.

Research services cost of revenue was less than \$0.1 million for both the three months ended June 30, 2021 and 2020.

Gross Profit

	Three Months Ended June 30,					
	2021		2020		Change	
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Gross profit	\$ 4,580	14%	\$ 2,851	12%	\$ 1,729	61%

Gross profit increased \$1.7 million, or 61%, to \$4.6 million for the three months ended June 30, 2021 from \$2.9 million in the comparable period in 2020. The increase in gross profit was the result of the \$7.1 million increase in total revenue, offset, in part, by the \$5.3 million increase in total cost of revenue. The increase in revenue was principally driven by growth in the petrochemical and refinery markets, particularly in the United States, and growth in the European building materials market, offset, in part, by a decrease in project-based demand in the subsea and LNG markets. The increase in total cost of revenue was the result of the \$2.5 million increase in material costs associated with 2.6 million square feet, or 35%, increase in total product shipments and the \$2.8 million increase in manufacturing expense during 2021.

Gross profit as a percentage of total revenue increased to 14% of total revenue for the three months ended June 30, 2021 from 12% in the comparable period in 2020.

Research and Development Expenses

	Three Months Ended June 30,				Change	
	2021		2020			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Research and development expenses	\$ 2,609	8%	\$ 2,121	9%	\$ 488	23%

Research and development expenses increased \$0.5 million, or 23%, to \$2.6 million for the three months ended June 30, 2021 from \$2.1 million in the comparable period in 2020. The \$0.5 million increase reflects an increase in compensation expense associated with the hiring of new employees to facilitate our efforts to optimize our carbon aerogel battery materials and develop next-generation aerogel compositions, form factors and manufacturing technologies.

Research and development expenses as a percentage of total revenue decreased to 8% for the three months ended June 30, 2021 from 9% in the comparable period in 2020 as a result of revenue growth in the three months ended June 30, 2021.

Sales and Marketing Expenses

	Three Months Ended June 30,				Change	
	2021		2020		Amount	Percentage
	Amount	Percentage of Revenue	Amount	Percentage of Revenue		
	(\$ in thousands)					
Sales and marketing expenses	\$ 3,568	11%	\$ 2,972	12%	\$ 596	20%

Sales and marketing expenses increased by \$0.6 million, or 20%, to \$3.6 million for the three months ended June 30, 2021 from \$3.0 million in the comparable period in 2020. The \$0.6 million increase was the result of increases in compensation and related costs of \$0.7 million and other sales related costs of \$0.2, offset, in part, by a decrease in sales consultant expenses of \$0.3 million.

Sales and marketing expenses as a percentage of total revenue decreased to 11% for the three months ended June 30, 2021 from 12% in the comparable period in 2020 as a result of revenue growth in the three months ended June 30, 2021.

General and Administrative Expenses

	Three Months Ended June 30,				Change	
	2021		2020		Amount	Percentage
	Amount	Percentage of Revenue	Amount	Percentage of Revenue		
	(\$ in thousands)					
General and administrative expenses	\$ 5,017	16%	\$ 3,406	14%	\$ 1,611	47%

General and administrative expenses increased by \$1.6 million, or 47%, to \$5.0 million for the three months ended June 30, 2021 from \$3.4 million in the comparable period in 2020. The \$1.6 million increase was the result of increases in professional services expenses of \$0.9 million, compensation and related costs of \$0.5 million, patent enforcement costs of \$0.1 million and other general and administrative expenses of \$0.2 million, offset, in part, by a decrease in the provision for uncollectible accounts of \$0.1 million.

General and administrative expenses as a percentage of total revenue increased to 16% for the three months ended June 30, 2021 from 14% in the comparable period in 2020.

Interest Expense, net

	Three Months Ended June 30,				Change	
	2021		2020		Amount	Percentage
	Amount	Percentage of Revenue	Amount	Percentage of Revenue		
	(\$ in thousands)					
Interest expense, net	\$ (55)	0%	\$ (50)	0%	\$ (5)	10%

Interest expense, net, consists primarily of fees and interest expense associated with our revolving credit agreement and was less than \$0.1 million for both the three months ended June 30, 2021 and 2020.

Six months ended June 30, 2021 compared to the six months ended June 30, 2020

The following tables set forth a comparison of the components of our results of operations for the periods presented:

Revenue

	Six Months Ended June 30,				Change	
	2021		2020			
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
(\$ in thousands)						
Revenue:						
Product	\$ 59,546	100%	\$ 52,833	100%	\$ 6,713	13%
Research services	221	0%	227	0%	(6)	(3)%
Total revenue	<u>\$ 59,767</u>	<u>100%</u>	<u>\$ 53,060</u>	<u>100%</u>	<u>\$ 6,707</u>	<u>13%</u>

The following chart sets forth product shipments in square feet for the periods presented:

	Six Months Ended June 30,		Change	
	2021	2020	Amount	Percentage
Product shipments in square feet (in thousands)	18,514	15,482	3,032	20%

Total revenue increased \$6.7 million, or 13%, to \$59.8 million for the six months ended June 30, 2021 from \$53.1 million in the comparable period in 2020. The increase in total revenue was principally the result of an increase in product revenue.

Product revenue increased by \$6.7 million, or 13%, to \$59.5 million for the six months ended June 30, 2021 from \$52.8 million in the comparable period in 2020. This increase was principally driven by growth in the petrochemical and refinery markets, particularly in the United States, growth in the European building materials market and an increase in project-based demand in the LNG market, offset, in part, by a decrease in project-based revenue in the subsea market.

Product revenue for the six months ended June 30, 2021 included \$17.0 million to a North American distributor and \$8.8 million to a European LNG project contractor. Product revenue for the six months ended June 30, 2020 included \$10.1 million to an Asian LNG project contractor and \$9.4 million to a North American distributor.

The average selling price per square foot of our products decreased by \$0.19, or 6%, to \$3.22 per square foot for the six months ended June 30, 2021 from \$3.41 per square foot for the six months ended June 30, 2020. The decrease in average selling price principally reflected the impact of a change in the mix of products sold for the six months ended June 30, 2021 from the comparable period in 2020. This decrease in average selling price had the effect of decreasing product revenue by \$3.6 million for the six months ended June 30, 2021 from the comparable period in 2020.

In volume terms, product shipments increased by 3.0 million square feet, or 20%, to 18.5 million square feet of aerogel products for the six months ended June 30, 2021, as compared to 15.5 million square feet for the six months ended June 30, 2020. The increase in product volume had the effect of increasing product revenue by \$10.3 million for the six months ended June 30, 2021 from the comparable period in 2020.

Research services revenue was \$0.2 million for both the six months ended June 30, 2021 and June 30, 2020.

Product revenue was nearly 100% of total revenue for the six months ended June 30, 2021 and June 30, 2020. Research services revenue was less than 1% of total revenue for the six months ended June 30, 2021 and June 30, 2020.

Cost of Revenue

	Six Months Ended June 30,						Change	
	2021			2020				
	Amount	Percentage of Related Revenue	Percentage of Total Revenue	Amount	Percentage of Related Revenue	Percentage of Total Revenue	Amount	Percentage
(\$ in thousands)								
Cost of revenue:								
Product	\$ 51,180	86%	86%	\$ 44,160	84%	83%	\$ 7,020	16%
Research services	51	23%	0%	69	30%	0%	(18)	(26)%
Total cost of revenue	\$ 51,231	86%	86%	\$ 44,229	83%	83%	\$ 7,002	16%

Total cost of revenue increased \$7.0 million, or 16%, to \$51.2 million for the six months ended June 30, 2021 from \$44.2 million in the comparable period in 2020. The increase in total cost of revenue was principally the result of an increase in product cost of revenue.

Product cost of revenue increased \$7.0 million, or 16%, to \$51.2 million for the six months ended June 30, 2021 from \$44.2 million in the comparable period in 2020. The \$7.0 million increase was the result of a \$4.3 million increase in material costs and a \$2.7 million increase in manufacturing expense. The increase in material costs was driven principally by the 3.0 million square feet, or 20%, increase in product shipments. The increase in manufacturing expense was the result of increases in compensation and related expenses of \$1.6 million, operating supplies of \$0.6 million, maintenance costs of \$0.4 million and other manufacturing expenses of \$0.2 million, partially offset by a decrease in professional services expenses of \$0.1 million.

Product cost of revenue as a percentage of product revenue increased to 86% during the six months ended June 30, 2021 from 84% during the six months ended June 30, 2020. This increase was the result of the increase in material costs as a percentage of revenue during the six months ended June 30, 2021.

Research services cost of revenue, consists primarily of costs associated with our fulfillment of our research services contracts and was less than \$0.1 million for both the six months ended June 30, 2021 and 2020.

Gross Profit

	Six Months Ended June 30,					
	2021		2020		Change	
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Gross profit	\$ 8,536	14%	\$ 8,831	17%	\$ (295)	(3)%

Gross profit decreased \$0.3 million, or 3%, to \$8.5 million for the six months ended June 30, 2021 from \$8.8 million in the comparable period in 2020. The decrease in gross profit was the result of the \$7.0 increase in total cost of revenue, offset, in part, by the \$6.7 million increase in total revenue. The increase in revenue was driven principally by the 20% increase in product shipments. The increase in total cost of revenue was driven by the \$4.3 million increase in material costs and the \$2.7 million increase in manufacturing expense.

Gross profit as a percentage of total revenue decreased to 14% of total revenue for the six months ended June 30, 2021 from 17% in the comparable period in 2020.

Research and Development Expenses

	Six Months Ended June 30,					
	2021		2020		Change	
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Research and development expenses	\$ 5,051	8%	\$ 4,348	8%	\$ 703	16%

Research and development expenses increased \$0.7 million, or 16%, to \$5.0 million for the six months ended June 30, 2021 from \$4.3 million in the comparable period in 2020. The \$0.7 million increase was the result of increases in compensation and related costs of \$0.6 million and other research and development expenses of \$0.1 million.

Research and development expenses as a percentage of total revenue was 8% for both the six months ended June 30, 2021 and 2020.

Sales and Marketing Expenses

	Six Months Ended June 30,					
	2021		2020		Change	
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
Sales and marketing expenses	\$ 6,869	11%	\$ 6,296	12%	\$ 573	9%

Sales and marketing expenses increased by \$0.6 million, or 9%, to \$6.9 million for the six months ended June 30, 2021 from \$6.3 million in the comparable period in 2020. The \$0.6 million increase was the result of increases in compensation and related expenses of \$1.1 million and other sales related expenses of \$0.2 million, offset, in part, by decreases in sales consultant costs of \$0.4 million and travel related expenses of \$0.3 million.

Sales and marketing expenses as a percentage of total revenue decreased to 11% for the six months ended June 30, 2021 from 12% for the six months ended June 30, 2020.

General and Administrative Expenses

	Six Months Ended June 30,					
	2021		2020		Change	
	Amount	Percentage of Revenue	Amount	Percentage of Revenue	Amount	Percentage
	(\$ in thousands)					
General and administrative expenses	\$ 9,405	16%	\$ 6,921	13%	\$ 2,484	36%

General and administrative expenses increased by \$2.5 million, or 36%, to \$9.4 million during the six months ended June 30, 2021 from \$6.9 million in the comparable period in 2020. The \$2.5 million increase was the result of an increase in compensation and related expenses of \$1.2 million, professional fees of \$1.2 million, and other general and administrative expenses of \$0.5 million, offset, in part, by decreases in the provision for uncollectible accounts of \$0.3 million and patent enforcement costs of \$0.1 million.

General and administrative expenses as a percentage of total revenue increased to 16% of total revenue for the six months ended June 30, 2021 from 13% during the comparable period in 2020.

Interest Expense, net

	Six Months Ended June 30,				Change	
	2021		2020		Amount	Percentage
	Amount	Percentage of Revenue	Amount	Percentage of Revenue		
	(\$ in thousands)					
Interest expense, net	\$ (130)	(0)%	\$ (133)	(0)%	\$ 3	(2)%

Interest expense, net, consists primarily of fees and interest expense associated with outstanding balances under our revolving credit agreement and was \$0.1 million during both the six months ended June 30, 2021 and 2020.

Liquidity and Capital Resources

Overview

We have experienced significant losses and invested substantial resources since our inception to develop, commercialize and protect our aerogel technology and to build a manufacturing infrastructure capable of supplying aerogel products at the volumes and costs required by our customers. These investments have included research and development and other operating expenses, capital expenditures and investment in working capital balances.

Through 2015, we experienced revenue growth and gained share in our target markets. Despite a decline in revenue in 2016, 2017 and 2018, our financial projections anticipated long-term revenue growth, increasing levels of gross profit and improved cash flow from operations. To support this growth, we initiated a plan in 2018 to increase the capacity of our East Providence, Rhode Island manufacturing facility to approximately 60 million square feet of aerogel blankets and currently expect to achieve this goal by the end of 2021. We may incur additional capital expenditures to complete this plan in 2021.

We are also increasing our investment in the research and development of next-generation aerogel products and technologies. During 2021, we will continue to develop aerogel products and technologies for the electric vehicle market. We believe the commercial potential for our technology in the electric vehicle market is significant. Accordingly, we are planning to hire additional personnel, incur additional operating expenses, build an automated thermal barrier fabrication operation, and construct a carbon aerogel battery materials facility, among other items.

In addition, we anticipate that we will need to increase our silica aerogel blanket manufacturing capacity to keep pace with the significant potential demand for our PyroThin thermal barriers. Accordingly, we are planning a significant expansion of our aerogel capacity prior to the end of 2023. The expected elements of the completed expansion plan will include the size of the required capacity expansion, the selection of an optimal manufacturing site for the expansion and a detailed timeline for the construction and operation of the facility. We expect that we will incur a significant increase in capital expenditures to build out the additional capacity and in operating expenses associated with the start-up of the facility.

We took several actions during 2020 to increase the financial resources available to support current operating requirements and capital expenditures. In February 2020, we completed an underwritten public offering of our common stock and received net proceeds of \$14.8 million. In March 2020, we extended the maturity of our revolving credit facility with Silicon Valley Bank to April 28, 2021. In May 2020, our wholly owned subsidiary, Aspen Aerogels Rhode Island, LLC, received PPP Loan proceeds of \$3.7 million under the CARES Act. During November and December 2020, we also completed the sale of 714,357 shares of our common stock at an average price of \$13.96 per share through our at-the-market (ATM) offering program with B. Riley Securities, Inc. as our sales agent and received net proceeds of \$9.5 million after deducting commissions \$0.3 million and offering expenses of approximately \$0.2 million.

During the six months ended June 30, 2021, we sold an additional 899,981 shares of our common stock at an average price of \$21.32 through the ATM offering program and received net proceeds of \$18.6 million. In addition, on June 29, 2021, we completed the Private Placement. We received net proceeds of \$73.6 million from the Private Placement after deducting fees and offering expenses of \$1.4 million.

We believe that our existing cash balance and funds available under our revolving credit facility will be sufficient to support current operating requirements, current research and development activities and the initial capital expenditures required to support the evolving commercial opportunity in the electric vehicle market and other strategic business opportunities.

However, we plan to supplement our cash balance and available credit with equity financings, debt financings, customer prepayments, or technology licensing fees to provide the additional capital necessary to complete future capacity expansions or to support evolving strategic business initiatives. We also intend to extend or replace our revolving credit facility with Silicon Valley Bank prior to its maturity.

Primary Sources of Liquidity

Our principal sources of liquidity are currently our cash and cash equivalents and our revolving credit facility with Silicon Valley Bank. Cash and cash equivalents consist primarily of cash and money market accounts on deposit with banks. As of June 30, 2021, we had \$102.3 million of cash and cash equivalents.

On February 18, 2020, we completed an underwritten public offering of 1,955,000 shares of our common stock at an offering price of \$8.25 per share. We received net proceeds of \$14.8 million after deducting underwriting discounts and commissions of \$1.1 million and offering expenses of approximately \$0.3 million.

On November 5, 2020, we entered into a sales agreement for an ATM offering program under which we may sell up to \$33,871,250 of our common stock through B. Riley Securities as our sales agent. We are not obligated to sell any stock under the sales agreement. We will pay B. Riley Securities a commission of 3.0% of the gross sales proceeds of shares sold under the agreement. During 2020, we sold 714,357 shares of our common stock through the ATM offering program and received net proceeds of \$9.5 million. During the six months ended June 30, 2021, we sold an additional 899,981 shares of our common stock through the ATM offering program and received net proceeds of \$18.6 million.

On June 29, 2021, we completed the Private Placement. We received net proceeds of \$73.6 million after deducting fees and offering expenses of \$1.4 million.

On May 1, 2020, our wholly-owned subsidiary, Aspen Aerogels Rhode Island, LLC (Borrower) executed a note for a loan of \$3.7 million pursuant to the PPP under the CARES Act, as amended, and administered by the SBA. The loan is unsecured, contains customary terms, including events of default, carries an interest rate of 1% per year, and matures on May 1, 2022. The Borrower may repay the loan in full at any time without penalty. In addition, the Borrower may apply to have the maturity of loan extended to May 1, 2025.

The Borrower may apply to have the PPP Loan indebtedness forgiven in whole or in part subject to SBA guidelines and based on the use of loan proceeds for payroll costs, mortgage interest payments, rent and utility costs over either an eight-week or 24-week period, at the Borrower's option, following its receipt of the loan proceeds. The SBA may deny the Borrower's loan forgiveness application if the agency determines that the Borrower was ineligible for the PPP Loan. As of June 30, 2021, the Borrower had not applied for forgiveness.

If the Borrower applies for, but SBA denies forgiveness of the PPP Loan in whole or in part, the Borrower will be required to make payments of the remaining principal and accrued interest in equal monthly installments over the remaining term of the loan. If the Borrower does not apply for forgiveness by August 19, 2021, the Borrower will be required to make payments of principal and accrued interest in equal monthly installments over the remaining term of the loan.

We have maintained our revolving credit facility, as amended from time to time, with Silicon Valley Bank since March 2011. On March 12, 2021, we amended and restated our revolving credit facility with Silicon Valley Bank to extend the maturity date of the revolving credit facility to April 28, 2022 and to establish certain minimum Adjusted EBITDA and minimum Adjusted Quick Ratio covenants. We intend to extend or replace the facility prior to its maturity.

Under our revolving credit facility, we may borrow a maximum of \$20.0 million, subject to continued covenant compliance and borrowing base requirements. The interest rate applicable to borrowings under the revolving credit facility is based on the prime rate,

as defined, subject to a minimum rate of 4.00% per annum. The rates applicable to borrowings vary from prime rate plus 0.75% per annum to prime rate plus 2.00% per annum. In addition, we are required to pay a monthly unused revolving line facility fee of 0.50% per annum of the average unused portion of the revolving credit facility.

At June 30, 2021, we had no outstanding borrowings under our revolving credit facility, \$1.5 million of outstanding letters of credit secured by the revolving credit facility, \$3.7 million outstanding on the PPP Loan, and an obligation of \$9.7 million associated with prepayments received pursuant to our supply agreement with BASF.

Under the revolving credit facility, we are required to comply with both non-financial and financial covenants, including minimum Adjusted EBITDA and Adjusted Quick Ratio covenants, as defined in the loan agreement. At June 30, 2021, we were in compliance with all such covenants.

The amount available to us under the revolving credit facility at June 30, 2021 was \$7.9 million after giving effect to the \$1.5 million of letters of credit outstanding.

Analysis of Cash Flow

Net Cash Used in Operating Activities

During the six months ended June 30, 2021, we used \$0.5 million in net cash in operating activities, as compared to the use of \$3.2 million in net cash during the comparable period in 2020, a decrease in the use of cash of \$2.7 million. This decrease in use of cash was the result of a decrease in net cash used by changes in operating assets and liabilities of \$7.1 million, offset, in part, by an increase in net loss adjusted for non-cash items of \$4.4 million.

Net Cash Used in Investing Activities

Net cash used in investing activities is primarily related to capital expenditures to support our growth. Net cash used in investing activities for the six months ended June 30, 2021 and 2020 was \$3.9 million and \$2.0 million, respectively, for capital expenditures primarily for machinery and equipment to improve the capacity, throughput, efficiency and reliability of our East Providence manufacturing facility.

Net Cash Provided by Financing Activities

Net cash provided by financing activities for the six months ended June 30, 2021 totaled \$90.2 million and consisted of \$73.6 million in net proceeds from the Private Placement, \$18.6 million in net proceeds from the ATM offering program, and \$0.7 million in proceeds from employee stock option exercises, offset, in part, by \$2.7 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units.

Net cash provided by financing activities for the six months ended June 30, 2020 totaled \$14.9 million and consisted of \$19.4 million in borrowings under our line of credit, \$14.8 million in net proceeds from an underwritten public offering of our common stock, \$3.7 million in net proceeds from the issuance of long-term debt, and \$0.9 million in proceeds from employee stock option exercises, offset, in part, by \$22.6 million of repayments under our line of credit and \$1.3 million in cash used for payments made for employee tax withholdings associated with the vesting of restricted stock units.

Off Balance Sheet Arrangements

Since inception, we have not engaged in any off balance sheet activities as defined in Item 303(a)(4) of Regulation S-K.

Contractual Obligations and Commitments

There have been no material changes to our contractual obligations and commitments as reported in our Annual Report.

Recent Accounting Pronouncements

Information regarding new accounting pronouncements is included in note 2 to our unaudited consolidated financial statements contained in Item 1 of this Quarterly Report on Form 10-Q.

Critical Accounting Policies and Estimates

Our financial statements are prepared in accordance with U.S. GAAP. The preparation of our financial statements and related disclosures requires us to make estimates, assumptions and judgments that affect the reported amount of assets, liabilities, revenue, costs and expenses and related disclosures. We believe that the estimates, assumptions and judgments involved in these accounting policies have the greatest potential impact on our financial statements and, therefore, we consider these to be our critical accounting policies. Accordingly, we evaluate our estimates and assumptions on an ongoing basis. Our actual results may differ from these estimates under different assumptions and conditions. See our Annual Report and note 2 to our consolidated financial statements included elsewhere in this Quarterly Report on Form 10-Q for information about these critical accounting policies, as well as a description of our other significant accounting policies.

Certain Factors That May Affect Future Results of Operations

The SEC encourages companies to disclose forward-looking information so that investors can better understand a company's future prospects and make informed investment decisions. This Quarterly Report on Form 10-Q contains such "forward-looking statements" within the meaning of the Private Securities Litigation Reform Act of 1995. These statements involve known and unknown risks, uncertainties and other important factors, which may cause our actual results, performance or achievements to be materially different from any future results, performances or achievements expressed or implied by the forward-looking statements. Forward-looking statements include, but are not limited to, statements about: our beliefs in the appropriateness of our assumptions, the accuracy of our estimates regarding expenses, loss contingencies, future revenues, future profits, uses of cash, available credit, PPP Loan Proceeds, capital requirements, and the need for additional financing to operate our business, including to complete the planned capacity expansion of our East Providence manufacturing facility, and to fund our planned strategic business initiatives; the performance of our aerogel blankets; our expectation that we will be successful in obtaining, enforcing and defending our patents against competitors and that such patents are valid and enforceable; our belief that our products possess strong competitive advantages over traditional insulation materials, including the superior thermal performance and the thin, easy-to-use and durable blanket form of our products; our plans to expand capacity in our East Providence, Rhode Island manufacturing facility; our estimates of annual production capacity; our expectation to achieve our EP20 goals by the end of 2021; our plans regarding the future capacity expansion, including the selection of a manufacturing site and the construction and operation of the facility; beliefs about the role of our technology and products in the electric vehicle market; beliefs about the commercial potential for our technology in the electric vehicle market; beliefs about our ability to produce and deliver products to electric vehicle customers; beliefs about Aspen's contracts with the major U.S. automotive manufacturer; beliefs about the potential for the major U.S. automotive manufacturer to become a significant customer for Aspen's products; beliefs about revenue, costs, expenses, profitability, investments or cash flow associated with the contracts with the major U.S. automotive manufacturer, beliefs about the performance of our thermal barrier products in the battery systems of electric vehicles; beliefs about the potential the commercial opportunity for Aspen's thermal barrier products; our strategic partnership with BASF and the potential benefits of such a relationship, including the potential for it to create new product and market opportunities; our supply agreement with BASF, our supply to BASF of its Spaceloft A2 product and newly developed product, the potential for future cash advances from BASF under our supply agreement with BASF (payment of which are subject to certain conditions) to provide a source of financing and the potential for BASF to become a significant customer for our products; our joint development agreement with BASF, and the potential for it to support the development of new aerogel products and technologies; our beliefs about the usefulness of the square foot operating metric; our beliefs about the financial metrics that are indicative of our core performance; our beliefs about the usefulness of our presentation of Adjusted EBITDA; our expectations about the effect of manufacturing capacity on financial metrics such as Adjusted EBITDA; our expectations about future revenues, expenses, gross profit, net loss, loss per share and Adjusted EBITDA, sources and uses of cash, capital requirements and the sufficiency of our existing cash balance and available credit; our beliefs about the outcome, effects or estimated costs of current or potential litigation or their respective timing, including expected legal expense in connection with our patent enforcement actions; our plans to devote substantial resources to the development of new aerogel technology; our expectations about product mix; our expectations about future material costs and manufacturing expenses as a percentage of revenue; our expectations of future gross profit and the effect of manufacturing expenses, manufacturing capacity and productivity on gross profit; our expectations about our resources and other investments in new technology and related research and development activities and associated expenses; our expectations about short and long term (a) research and development (b) general and administrative and (c) sales and marketing expenses; our expectations of revenue growth, increased gross profit, and improving cash flows over the long term; our intentions about managing capital expenditures and working capital balances; our expectations about incurring significant capital expenditures in the future; our expectations about the expansion of our workforce and resources and its effect on sales and marketing, general and

administrative, and related expenses; our expectations about future product revenue and demand for our products; our expectations about the effect of stock based compensation on various costs and expenses; our expectations about potential sources of future financing; our beliefs about the impact of accounting policies on our financial statements; our beliefs about the effect of interest rates, inflation and foreign currency fluctuations on our results of operations and financial condition; our beliefs about the expansion of our international operations; our statements about the impact of major public health concerns, including the COVID-19 pandemic or other pandemics arising globally, and the future, and currently unknown extent of, the impact of the COVID-19 pandemic on our business and operations; our statements about the sufficiency of our current and future actions to address the impact of the COVID-19 pandemic on our business and operations, including our future revenue, Adjusted EBITDA and other financial metrics; our belief that we qualify for partial or complete forgiveness of the PPP Loan; and changes by governmental authorities regarding the CARES Act or related administrative matters and the Company's and its subsidiary's abilities to comply with the terms of the PPP Loan and the CARES Act, including to use the proceeds of the PPP Loan as described herein.

Words such as “may,” “will,” “anticipate,” “estimate,” “expects,” “projects,” “intends,” “plans,” “believes” and words and terms of similar substance used in connection with any discussion of future operating or financial performance, identify forward-looking statements. All forward-looking statements are management's present expectations of future events and are subject to a number of risks and uncertainties that could cause actual results to differ materially and adversely from those described in the forward-looking statements. These risks include, but are not limited to, those set forth in this Quarterly Report on Form 10-Q and under the heading “Risk Factors” contained in Item 1A of our Annual Report.

In light of these assumptions, risks and uncertainties, the results and events discussed in the forward-looking statements contained in this Quarterly Report on Form 10-Q might not occur. Stockholders and other readers are cautioned not to place undue reliance on the forward-looking statements, which speak only as of the date of this Quarterly Report on Form 10-Q. We are not under any obligation, and we expressly disclaim any obligation, to update or alter any forward-looking statements, whether as a result of new information, future events or otherwise. All subsequent forward-looking statements attributable to Aspen Aerogels, Inc. or to any person acting on its behalf are expressly qualified in their entirety by the cautionary statements contained or referred to in this section.

Item 3. Quantitative and Qualitative Disclosures About Market Risk.

Market risk represents the risk of loss that may impact our financial position due to adverse changes in financial market prices and rates. Our market risk exposure results primarily from fluctuations in interest rates as well as from inflation. In the normal course of business, we are exposed to market risks, including changes in interest rates which affect our line of credit under our revolving credit facility as well as cash flows. We may also face additional exchange rate risk in the future as we expand our business internationally.

Interest Rate Risk

We are exposed to changes in interest rates in the normal course of our business. At June 30, 2021, we had unrestricted cash and cash equivalents of \$102.3 million. These amounts were held for working capital and capital expansion purposes and were invested primarily in deposit and money market accounts at a major financial institution in North America. Due to the short-term nature of these investments, we believe that our exposure to changes in the fair value of our cash as a result of changes in interest rates is not material.

As of June 30, 2021, we had no borrowings outstanding on our revolving credit facility. At June 30, 2021, we had \$1.5 million of outstanding letters of credit supported by the revolving credit facility.

Under our revolving credit facility, we are permitted to borrow a maximum of \$20.0 million, subject to continued covenant compliance and borrowing base requirements. The interest rate applicable to borrowings under the revolving credit facility is based on the prime rate, as defined, subject to a minimum rate of 4.00% per annum. The rates applicable to borrowings vary from prime rate plus 0.75% per annum to prime rate plus 2.00% per annum. In addition, we are required to pay a monthly unused revolving line of credit facility fee of 0.5% per annum of the average unused portion of the revolving credit facility. The maturity date of our revolving credit facility is April 28, 2022. We intend to extend or replace the facility prior to its maturity.

At June 30, 2021, the amount available to us under the revolving credit facility was \$7.9 million after giving effect to the \$1.5 million of letters of credit outstanding under the facility.

Our PPP Loan has an interest rate of 1% per annum and matures on May 1, 2022. In accordance with the Flexibility Act, we may submit an application to extend the maturity of the loan by three years to May 1, 2025. At June 30, 2021, the PPP Loan had an outstanding balance of \$3.7 million.

Inflation Risk

Although we expect that our operating results will be influenced by general economic conditions, we do not believe that inflation has had a material effect on our results of operations during the periods presented in this report. However, our business may be affected by inflation in the future.

Foreign Currency Exchange Risk

We are subject to inherent risks attributed to operating in a global economy. Principally all of our revenue, receivables, purchases and debts are denominated in U.S. dollars.

Item 4. Controls and Procedures.

(a) Evaluation of Disclosure Controls and Procedures.

We maintain disclosure controls and procedures that are designed to ensure that information required to be disclosed in the reports that we file or submit under the Securities Exchange Act of 1934, as amended (the Exchange Act), is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure.

As of June 30, 2021, our management, with the participation of our principal executive officer and principal financial officer, evaluated the effectiveness of our disclosure controls and procedures (as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act). Our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving their objectives, and management necessarily applies its judgment in evaluating the cost-benefit relationship of possible controls and procedures. Based on such evaluation, our principal executive officer and principal financial officer have concluded that, as of June 30, 2021, our disclosure controls and procedures were effective to ensure that information required to be disclosed by us in the reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms, and is accumulated and communicated to our management, including our principal executive officer and principal financial officer, or persons performing similar functions, as appropriate to allow timely decisions regarding required disclosure. In addition, our principal executive officer and principal financial officer have concluded that the impact of the COVID-19 pandemic did not impact our ability to maintain our internal controls over financial reporting and disclosure controls and procedures.

(b) Changes in Internal Controls.

During the six months ended June 30, 2021, there were no changes in our internal control over financial reporting, as such term is defined in Rules 13a-15(f) and 15d-15(f) promulgated under the Exchange Act, that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

PART II — OTHER INFORMATION

Item 1. Legal Proceedings.

Patent Enforcement Actions

In May 2016, we filed a complaint for patent infringement against Nano Tech Co., Ltd. (Nano), and Guangdong Alison Hi Tech., Ltd. (Alison) in the International Trade Commission, or ITC. In February 2018, the ITC issued its final determination that Nano and Alison had infringed asserted Aspen patents and that they have not proven the patents are invalid except with respect to one dependent product claim, which the ITC found was not infringed. The ITC affirmed that Alison and Nano each violated Section 337 of the Tariff Act and issued a limited exclusion order prohibiting importation of infringing aerogel insulation products manufactured by Alison and Nano. Alison's appeal with respect to a product patent to the United States Court of Appeals for the Federal Circuit (CAFC) was rejected, and resulted in CAFC affirming the validity of our patent. The exclusion order, which is enforced by the United States Customs and Border Protection, is currently in effect.

Additionally, the USPTO denied Alison's requests to invalidate the claims of four of our patents in Inter-Partes Review. Alison also filed multiple similar requests with the Chinese Patent Office (SIPO), seeking to invalidate our Chinese manufacturing process patents and two of our Chinese product patents. With respect to one of those requests, not withdrawn previously by Alison, the Patent Review Board of SIPO (PRB), issued a decision upholding the validity of Aspen's issued patent as amended in the proceedings. Alison has appealed the PRB's decision to the Beijing IP court. On July 25, 2020, the Beijing IP court dismissed Alison's appeal and upheld the validity of Aspen's patent and we received this decision on September 15, 2020. Nano has also filed a request seeking invalidation of a product patent at SIPO. After the oral hearing at PRB, Nano withdrew its invalidation request. On September 23, 2019, Alison filed yet another request to invalidate the same patent, whose validity was previously confirmed by PRB. On January 23, 2020 PRB denied Alison's latest invalidation request.

In April 2016, we also filed a patent infringement suit at the District Court in Mannheim, Germany (Mannheim court), against Nano, Alison and two European resellers asserting their infringement of one of our German patents. We subsequently asserted infringement of another three patents against Nano, Alison and a European reseller of Alison's products at the Mannheim court. We have since settled with one European reseller in exchange for a commitment not to procure infringing products and cooperation with our case.

- In January 2018, the court issued a series of judgments by acknowledgement (German, Anerkenntnisurteil) finding the second reseller, Hiltex, liable for infringement and also issued injunctions against Hiltex. The judgments resulted from a settlement agreement in which Hiltex agreed not to resell the infringing products in Europe where at least one of the asserted patents are active.
- On March 8, 2019, the Mannheim court issued two separate judgments in cases against Nano and Alison, respectively. The Mannheim court determined that both Nano and Alison are infringing on Aspen's EP1638750 (750 Patent) in connection with their respective products. The court also issued injunctions prohibiting the offer, putting on the market, using, importing or possessing the infringing products. The court found the defendants liable to us for damages since September 22, 2012. The court also ordered the defendants to provide information on the scope of the acts of infringement committed since August 22, 2012, and a recall of infringing products. The court ordered Nano and Alison to bear the costs of the legal proceedings and reimburse statutory attorneys' costs and expenses to us, that exact amount of which is yet to be determined. Nano and Alison have appealed the judgments of the Mannheim court. Nano subsequently withdrew the appeal while Alison's appeal is currently pending.
- The Mannheim court issued two decisions on December 23, 2019 finding that Alison infringed the 577 Patent and the 950 Patent and also issued injunctions prohibiting Alison from continuing infringement in connection with any aerogel sheets. The December 2019 decisions against Alison have now become final and binding.
- The Mannheim court issued two decisions on July 31, 2020 finding that Nano infringed each of the 577 Patent and the 950 Patent. In addition to granting other remedies, the court also issued injunctions prohibiting the offer, putting on the market, using, importing or possessing any aerogel sheets. After the passing of deadline to file appeals, these decisions have now become final.
- Nano and Alison also initiated nullity actions in German Federal Patent Court in Munich against our asserted German patents. On September 25, 2018, the Federal Patent Court in Munich dismissed the challenge to the validity of 750 Patent which has subsequently become final. Nano and Alison also filed an opposition to one of the asserted patents at the EPO. In December

2018, the opposition division of EPO determined the patent, EP2813338 (338 Patent), was invalid on formality grounds and decided to revoke it, which determination is currently under appeal at the EPO Board of appeals.

- On March 19 and 20, 2019 the German Federal Patent Court in Munich (FPC) conducted oral proceedings and voided four claims in EP2415577 (577 Patent) and confirmed the validity of challenged claims in EP2422950 (950 Patent) within the scope of silica gels. These FPC judgments are now final and binding on the parties. Nano had filed another nullity action seeking to invalidate the remaining claims in the 577 Patent which action Nano subsequently failed to pursue. On June 17, 2020, Nano also filed an opposition to a recently issued Aspen Patent EP3120983B1, titled “Continuous Sheet of Gel Materials and Continuous sheet of Aerogel”.
- On January 28, 2021, a search order was executed and relevant evidence secured at the principal places of business of AMA S.p.A. and AMA Composites S.r.l. (collectively, AMA) in San Martino in Rio and Campogalliano, respectively, based on an ex-parte search order issued by the Court of Genoa, Italy at our request in connection with alleged infringement of the Italian part of our patents previously asserted successfully against Nano and Alison in Germany. The Court of Genoa subsequently held a hearing and confirmed the validity of the search order and its execution. While the search proceedings do not take a position on the infringement issues, we may use any evidence collected during the search proceedings to prove infringement. As a result, on May 3, 2021, we filed an infringement complaint, a writ of summons, as known in Italy, at the Court of Genoa alleging that AMA has infringed the Italian part of three European patents (same patents asserted in the German litigation) and a patent on composition of aerogel-based composites in connection with AMA’s resale of aerogel products supplied by Chinese companies and sale of any products derived therefrom. We are seeking monetary damages and preliminary injunction of AMA’s alleged infringing activities. We expect the Court of Genoa to assess our claims and AMA’s defense through appointment of an expert after the submission of relevant writs and evidence. We issued a press release on May 6, 2021 describing the patent enforcement action of May 3, 2021 (Press Release). On June 7, 2021, AMA served us a copy of a request it previously filed with the Court of Genoa seeking an ex-parte preliminary injunction (PI) against us alleging the Press Release constituted anti-competitive conduct and that it infringed AMA’s trademark rights. The service of the request followed the court’s prior denial of the ex-parte order and an order requiring AMA to serve the request on us. The court subsequently conducted an oral hearing on June 15, 2021. On June 24, 2021, the court denied AMA’s request for a PI, reasoned that Aspen’s Press Release was factually accurate, was not misleading, distinguished facts from opinions and that it was neither anti-competitive nor did it infringe trademark rights of AMA. The Court also ordered AMA to pay certain of our legal fees. AMA had until July 8, 2021 to appeal the denial of the PI to a panel of judges at the same court. On July 5, 2021, AMA informed us that it has decided not to appeal the denial of June 24, 2021. The denial of the PI in interim proceedings does not prevent AMA from bringing similar claims in patent infringement proceedings on the merits as a counter-claim.

Additionally, a reseller of Nano’s products in Taiwan challenged the validity of one of our patents in Taiwan in 2018. After careful review of our written response, the Taiwanese patent office has determined the patent as valid and dismissed the challenge in December 2018. In 2018, LG Chem Ltd. challenged the validity of one of our patents in Korea at the IPTAB of the Korean Intellectual Property Office. After conducting an oral hearing, the IPTAB issued a decision on November 30, 2019 upholding claims related to aerogel sheets incorporating fibers. On January 14, 2021 the Korean Patent Court confirmed the validity of the claims related to aerogel sheets incorporating fibers.

Due to their nature, it is difficult to predict the outcome or the costs involved in any litigation or administrative proceedings, including any appeals process. Furthermore, the counter parties in these proceedings may have significant resources and interest to litigate and therefore, these litigation matters could be protracted and may ultimately involve significant legal expenses. In addition to the foregoing, we have been and may be from time to time a party to other legal proceedings that arise in the ordinary course of business and to other patent enforcement actions to assert our patent rights.

Item 1A. Risk Factors.

There have been no material changes to the risk factors included in our Annual Report.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds.

(a) Unregistered Sales of Equity Securities.

On June 29, 2021, we entered into a securities purchase agreement (the Purchase Agreement) with a certain institutional and accredited investor (the Purchaser). Pursuant to the terms of the Purchase Agreement, we agreed to sell to the Purchaser an aggregate of 3,462,124 shares of our common stock at a purchase price equal to \$21.663 per share, for aggregate gross proceeds of approximately \$75.0 million (the Offering). The Offering closed on June 30, 2021. The Offering is exempt from registration pursuant

to the exemption for transactions by an issuer not involving any public offering under Section 4(a)(2) the Securities Act and Regulation D under the Securities Act. The foregoing securities are deemed restricted securities for purposes of the Securities Act.

(b) *Use of Proceeds from Initial Public Offering of Common Stock.*

Not applicable.

(c) *Purchases of Equity Securities By the Issuer and Affiliated Purchasers.*

We did not repurchase any of our equity securities during the quarter ended June 30, 2021.

Item 3. Defaults Upon Senior Securities.

None.

Item 4. Mine Safety Disclosures.

Not applicable.

Item 5. Other Information.

None.

Item 6. Exhibits.

(a) Exhibits

- 10.1 [Securities Purchase Agreement, dated June 29, 2021, by and between the Company and the purchaser identified therein \(incorporated by reference to Exhibit 10.1 of the Registrant's Form 8-K, filed on June 29, 2021\).](#)
- 10.2 [Performance-Based Restricted Stock Agreement, dated as of June 29, 2021, by and between the Registrant and Donald R. Young +](#)
- 10.3 [Third Amendment to Executive Agreement, dated as of June 29, 2021, by and between the Registrant and Donald R. Young. +](#)
- 31.1 [Certification of principal executive officer under Section 302\(a\) of the Sarbanes-Oxley Act of 2002.](#)
- 31.2 [Certification of principal financial officer under Section 302\(a\) of the Sarbanes-Oxley Act of 2002.](#)
- 32 [Certifications of the principal executive officer and the principal financial officer under Section 906 of the Sarbanes-Oxley Act of 2002.](#)
- 101.INS XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document.
- 101.SCH Inline XBRL Taxonomy Extension Schema Document.
- 101.CAL Inline XBRL Taxonomy Extension Calculation Linkbase Document.
- 101.DEF Inline XBRL Taxonomy Extension Definition Linkbase Document.
- 101.LAB Inline XBRL Taxonomy Extension Label Linkbase Document.
- 101.PRE Inline XBRL Taxonomy Extension Presentation Linkbase Document.
- 104 Cover Page Interactive Data File (embedded within the Inline XBRL document).

+ Management contract or compensatory plan or arrangement.

SIGNATURES

Pursuant to the requirements of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

ASPEN AEROGELS, INC.

Date: August 4, 2021

By: /s/ Donald R. Young
Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Date: August 4, 2021

By: /s/ John F. Fairbanks
John F. Fairbanks
Vice President, Chief Financial Officer and Treasurer
(principal financial officer and principal accounting officer)

PERFORMANCE-BASED RESTRICTED STOCK AGREEMENT

ASPEN AEROGELS, INC.

AGREEMENT made as of the 29th day of June, 2021 (the “Grant Date”), between Aspen Aerogels, Inc. (the “Company”), a Delaware corporation having its principal place of business in Northborough, Massachusetts and Donald R. Young (the “Participant”).

WHEREAS, the Company has adopted the 2014 Employee, Director and Consultant Equity Incentive Plan (the “Plan”) to promote the interests of the Company by providing an incentive for Employees, directors and Consultants of the Company or its Affiliates;

WHEREAS, pursuant to the provisions of the Plan, the Company desires to offer to the Participant shares of the Company’s common stock, \$0.00001 par value per share (“Common Stock”), in accordance with the provisions of the Plan, all on the terms and conditions hereinafter set forth;

WHEREAS, the Participant wishes to accept said offer; and

WHEREAS, the parties hereto understand and agree that any terms used and not defined herein have the meanings ascribed to such terms in the Plan.

NOW, THEREFORE, in consideration of the premises and the mutual covenants contained herein and for other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the parties hereto hereby agree as follows:

1. Terms of Grant. The Participant hereby accepts the offer of the Company to issue to the Participant, in accordance with the terms of the Plan and this Agreement, 461,616 Shares of the Company’s Common Stock (such shares, subject to adjustment pursuant to Section 25 of the Plan and Subsection 2.1(g) hereof, the “Granted Shares”) at a purchase price per share of \$0.00001 (the “Purchase Price”), receipt of which is hereby acknowledged by the Company by the Participant’s prior service to the Company and which amount will be reported as income on the Participant’s W-2 (or other applicable form).

2.1. Forfeiture Provisions.

(a) Lapsing Forfeiture Right. In the event that for any reason (i) the Participant is no longer an employee, director or consultant of the Company or an Affiliate (such event being the “Termination”) prior to achievement of the applicable Share Price (as defined below) as set forth below; or (ii) the Share Price is not achieved as set forth below by the fifth (5th) anniversary of the Grant Date (the “Performance End Date”), the Participant (or the Participant’s Survivor) shall, on the date of Termination or the Performance End Date, as applicable, immediately forfeit to the Company (or its designee) the number of Granted Shares as to which the Company’s forfeiture right has not yet lapsed as set forth below (the “Lapsing Forfeiture Right”) except as otherwise set forth in Section 2.1(c) and Section 2.1(d). The Lapsing Forfeiture Right shall lapse with respect to the number of the Granted Shares set forth below upon achievement of the Share Price prior to the date of Termination or the Performance End Date, as applicable (The “Earned Shares”). The foregoing notwithstanding, if a Share Price goal is achieved prior to the second (2nd) anniversary of the Grant Date, then the Lapsing Forfeiture Right as to the number of Earned Shares applicable to such Share Price goal shall not lapse until the third (3rd) anniversary of the Grant Date, and if a Share Price goal is achieved after the second (2nd) anniversary of the Grant Date, then the Lapsing Forfeiture Right as to the number of Earned Shares applicable to such Share Price goal shall not lapse until the date that is one (1) year after the date that the applicable Share Price goal is achieved; provided that if the applicable Share Price goal is achieved, the Earned Shares applicable to such Share Price goal will vest no later than the day prior to the Performance End Date.

If a Termination occurs prior to the third (3rd) anniversary of the Grant Date or prior to the date that is one year after the date that the applicable Share Price goal is achieved, as applicable, the Earned Shares shall be forfeited to the Company as if the Share Price goal had not been achieved as of the Termination date; provided, however, that if such

Termination occurs due to the Participant's termination without Cause (as defined in the Participant's Employment Agreement with the Company dated as of January 1, 2019, and as amended August 1, 2019 and January 1, 2020 (the "Employment Agreement")), the Participant's resignation for Good Reason (as defined in the Employment Agreement), or the Participant's death or Disability (as defined in the Plan), the Earned Shares applicable to such Share Price goal shall not be forfeited to the Company, and the Lapsing Forfeiture Right with respect to such Earned Shares shall lapse as of such Termination date.

The Lapsing Forfeiture Right shall lapse with respect to number of the Granted Shares set forth below, and such shares shall become Earned Shares, upon the achievement of a stock price that equals or exceeds a price per share (adjusted for any stock splits or stock dividends after the grant date in accordance with Section 25 of the Plan) (the "Share Price") calculated as follows: (A) the 30 day volume weighted average price of the Company's common stock on a national securities exchange (the "30-Day VWAP") for at least 60 consecutive trading days; or (B) in event of a Change of Control Sale of the Company (as defined in Section 2.1(d) below), the per share value received by the Company's shareholders in the sale of the Company; provided that for purposes of calculating the per share value received by the Company's shareholders, cash consideration would be valued at par, any publicly traded securities received as consideration would be valued at their 60 day volume weighted average price (the "60-day VWAP") of the acquiring company's common stock on a national securities exchange as of the closing date of the Change of Control Sale of the Company, and any securities that are not publicly traded or for which a 60-day VWAP cannot be calculated will be valued by the Company's Board of Directors in its reasonable judgment (the Change of Control Vesting Price").

The Lapsing Forfeiture Right shall lapse with respect to:

- 34% of the Granted Shares upon the achievement of a Share Price that equals or exceeds \$43.33;
- 33% of the Granted Shares upon the achievement of a Share Price that equals or exceeds \$64.99; and
- 33% of the Granted Shares upon the achievement of a Share Price that equals or exceeds \$86.65.

Any Granted Shares as to which the Company's Lapsing Forfeiture Right has not previously lapsed upon achievement of the above Share Price goals shall be forfeited to the Company on the Performance End Date, or earlier upon a Termination, subject to the provisions of Sections 2.1(c) and (d) below,).

(b) Effect of a For Cause Termination or Resignation without Good Reason. Notwithstanding anything to the contrary contained in this Agreement or in the Participant's Employment Agreement, including without limitation, Section 5(a)(xi) of the Employment Agreement, in the event the Company terminates the Participant's service as an Employee, director or Consultant for Cause (as defined in the Employment Agreement) or in the event the Board of Directors determines, within one year after the Participant's termination, that either prior or subsequent to the Participant's termination the Participant engaged in conduct that would constitute Cause, or if the Participant resigns without Good Reason (as defined in the Employment Agreement), all of the Granted Shares then held by the Participant that have not vested shall be forfeited to the Company immediately as of the time the Participant is notified that he or she has been terminated for Cause or that he or she engaged in conduct which would constitute Cause or as of the effective time of the Participant's resignation without Good Reason.

(c) Effect of a Termination without Cause or Resignation for Good Reason. Notwithstanding anything to the contrary contained in this Agreement or the Employment Agreement, including, without limitation, Section 4(b)(vi) of the Employment Agreement, but in accordance with the requirement that the release referred to in Section 4(b) of the Employment Agreement (the "Release") be effective and irrevocable, in the event the Company terminates the Participant's service as an Employee, director or Consultant without Cause (as defined in the Employment Agreement) or if the Participant resigns for Good Reason (as defined in the Employment Agreement), the Company's Lapsing Forfeiture Right shall lapse with respect to a portion of the Granted Shares then held by the Participant as to which the Company's Lapsing Forfeiture Right has not previously lapsed, and such shares will become Earned Shares, as follows:

- If the 60-day VWAP of the Company's common stock on a national securities exchange on the Termination date (the "Termination VWAP") is less than \$43.33, the Company's Lapsing Forfeiture Right shall lapse with respect to 34% of the Granted Shares multiplied by a fraction determined as follows:
-

Termination VWAP - \$21.663

\$21.66

- If the Termination VWAP is greater than \$43.33 and less than \$64.99, the Company's Lapsing Forfeiture Right shall lapse with respect to 33% of the Granted Shares multiplied by a fraction determined as follows:

Termination VWAP - \$43.33

\$21.66

- If the Termination VWAP is greater than \$64.99 and less than \$86.65, the Company's Lapsing Forfeiture Right shall lapse with respect to 33% of the Granted Shares multiplied by a fraction determined as follows:

Termination VWAP - \$64.99

\$21.66

In addition, any unearned tranche of Granted Shares that would be earned in accordance with Section 2.1(a) if the Share Price was calculated as the Termination VWAP will become Earned Shares. Upon the Release becoming effective and irrevocable, all Earned Shares (including any shares earned as a result of the application of this Section 2.1(c)) shall become fully vested. The remainder of the Granted Shares held by the Participant as to which the Company's Lapsing Forfeiture Right has not lapsed shall be forfeited to the Company.

(d) Effect of Change of Control. Notwithstanding anything to the Contrary contained in the Employment Agreement, including Section 4(b)(vi) or Section 5(a)(xi) thereof, in addition to any vesting provided for in Section 2.1(a), the Company's Lapsing Repurchase Right shall terminate, and the Participant's ownership of a portion of the Granted Shares then held by the Participant, shall become vested in the event of a Change of Control Sale of the Company (as defined below) and such shares will become Earned Shares, in accordance with the following:

- If the Change of Control Vesting Price is less than \$43.33, the Company's Lapsing Forfeiture Right shall lapse with respect to 34% of the Granted Shares multiplied by a fraction determined as follows:

Change of Control Vesting Price - \$21.663

\$21.66

- If the Change of Control Vesting Price is greater than \$43.33 and less than \$64.99, the Company's Lapsing Forfeiture Right shall lapse with respect to 33% of the Granted Shares multiplied by a fraction determined as follows:

Change of Control Vesting Price - \$21.663

\$43.33

- If the Change of Control Vesting Price is greater than \$64.99 and less than \$86.65, the Company's Lapsing Forfeiture Right shall lapse with respect to 33% of the Granted Shares multiplied by a fraction determined as follows:

Change of Control Vesting Price - \$21.663

\$64.99

In addition, any unearned tranche of Granted Shares that would be earned in accordance with Section 2.1(a) if the Share Price was calculated as the Change of Control Vesting Price will become Earned Shares. Any shares that become Earned Shares as a result of the application of this Section 2.1(d), shall nonetheless remain subject to the time-based vesting requirements of Section 2.1(a), subject to the provisions of Section 2.1(c). Granted Shares held by the Participant as to which the Company's Lapsing Forfeiture Right has not lapsed shall be forfeited to the Company.

Notwithstanding the foregoing, in the event the successor company in the Change of Control Sale of the Company does not assume the Company's obligations under this Agreement, the Lapsing Forfeiture Right shall lapse as to all Earned Shares as of immediately prior to the closing of the Change of Control Sale of the Company.

"Change of Control Sale of the Company" means the sale of the Company in which proceeds are received by the Company's shareholders in connection with the occurrence of any of the following events:

- (i) **Ownership.** Any "Person" (as such term is used in Sections 13(d) and 14(d) of the Exchange Act) becomes the "Beneficial Owner" (as defined in Rule 13d-3 under the Exchange Act), directly or indirectly, of securities of the Company representing 50% or more of the total voting power represented by the Company's then outstanding voting securities (excluding for this purpose any such voting securities held by the Company or its Affiliates or by any employee benefit plan of the Company) pursuant to a transaction or a series of related transactions; or
- (ii) **Merger/Sale of Assets.** (A) A merger or consolidation of the Company whether or not approved by the Board of Directors, other than a merger or consolidation which would result in the voting securities of the Company outstanding immediately prior thereto continuing to represent (either by remaining outstanding or by being converted into voting securities of the surviving entity or the parent of such corporation) more than 50% of the total voting power represented by the voting securities of the Company or such surviving entity or parent of such corporation, as the case may be, outstanding immediately after such merger or consolidation; or (B) the sale or disposition by the Company of all or substantially all of the Company's assets in a transaction requiring stockholder approval; or
- (iii) **Change in Board Composition.** A change in the composition of the Board of Directors, as a result of which fewer than a majority of the directors are Incumbent Directors. "Incumbent Directors" shall mean directors who either (A) are directors of the Company as of the date hereof, or (B) are elected, or nominated for election, to the Board of Directors with the affirmative votes of at least a majority of the Incumbent Directors at the time of such election or nomination (but shall not include an individual whose election or nomination is in connection with an actual or threatened proxy contest relating to the election of directors to the Company).

(d) **Escrow.** The certificates representing all Granted Shares acquired by the Participant hereunder which from time to time are subject to the Lapsing Forfeiture Right shall be delivered to the Company and the Company shall hold such Granted Shares in escrow as provided in this Subsection 2.1(c). Upon the request of the Participant, the Company shall promptly release from escrow and deliver to the Participant the whole number of Granted Shares, if any, as to which the Company's Lapsing Forfeiture Right has lapsed and without the legend set forth in Section 5. In the event of forfeiture to the Company of Granted Shares subject to the Lapsing Forfeiture Right, the Company shall release from escrow and cancel a certificate for the number of Granted Shares so forfeited. Any cash or securities distributed in respect of the Granted Shares held in escrow, including, without limitation, ordinary cash dividends or shares issued as a result of stock splits, stock dividends or other recapitalizations ("Retained Distributions"), shall also be held in escrow in the same manner as the Granted Shares and all Retained Distributions shall be forfeited to the Company or released from escrow and delivered to the Participant, as the case may be, at such time and in such manner as the Granted Shares to which such Retained Distributions so relate. All ordinary cash dividends retained hereunder shall, during the period in which such dividends are retained by the Company, be deposited into an account at a financial institution selected by the Company, which shall not be required to bear interest or be segregated in a separate account.

(e) **Prohibition on Transfer.** The Participant recognizes and agrees that all Granted Shares and Retained Distributions which are subject to the Lapsing Forfeiture Right may not be sold, transferred, assigned, hypothecated, pledged, encumbered or otherwise disposed of, whether voluntarily or by operation of law, other than to the Company (or its designee). However, the Participant, with the approval of the Administrator, may transfer the Granted Shares and Retained Distributions for no consideration to or for the benefit of the Participant's Immediate Family (including, without limitation, to a trust for the benefit of the Participant's Immediate Family or to a partnership or limited liability company for one or more members of the Participant's Immediate Family), subject to such limits as the Administrator may establish,

and the transferee shall remain subject to all the terms and conditions applicable to this Agreement prior to such transfer and each such transferee shall so acknowledge in writing as a condition precedent to the effectiveness of such transfer. The term “Immediate Family” shall mean the Participant’s spouse, former spouse, parents, children, stepchildren, adoptive relationships, sisters, brothers, nieces and nephews and grandchildren and, for this purpose, shall also include the Participant. The Company shall not be required to transfer any Granted Shares or Retained Distributions on its books which shall have been sold, assigned or otherwise transferred in violation of this Subsection 2.1(d), or to treat as the owner of such Granted Shares or Retained Distributions, or to accord the right to vote as such owner or to pay dividends to, any person or organization to which any such Granted Shares or Retained Distributions shall have been so sold, assigned or otherwise transferred, in violation of this Subsection 2.1(d).

(f) Failure to Deliver Granted Shares to be Forfeited. In the event that the Granted Shares to be forfeited to the Company under this Agreement are not in the Company’s possession pursuant to Subsection 2.1(c) above or otherwise and the Participant or the Participant’s Survivor fails to deliver such Granted Shares to the Company (or its designee), the Company may immediately take such action as is appropriate to transfer record title of such Granted Shares from the Participant to the Company (or its designee) and treat the Participant and such Granted Shares in all respects as if delivery of such Granted Shares had been made as required by this Agreement. The Participant hereby irrevocably grants the Company a power of attorney which shall be coupled with an interest for the purpose of effectuating the preceding sentence.

(g) Adjustments. The Plan contains provisions covering the treatment of Shares in a number of contingencies such as stock splits and mergers. Provisions in the Plan for adjustment with respect to the Shares and the related provisions with respect to successors to the business of the Company are hereby made applicable hereunder and are incorporated herein by reference. These provisions are not intended by this Agreement to be made applicable to Company capital raising transactions.

(h) Clawback. In the event the Company determines it must restate its financial results as reported in a Form 10-K, Form 10-Q or other report filed with the Securities and Exchange Commission to correct an accounting error due to material noncompliance with any financial reporting requirement under the U. S. federal securities laws (a “Restatement”), and the Participant’s intentional misconduct during the period from the Grant Date to the fifth (5th) anniversary of the Grant Date caused or contributed to the need for the Restatement, the Company will seek to recover, at the direction of the Compensation Committee of the Board of Directors (the “Committee”) after it has reviewed the facts and circumstances that led to the requirement for the Restatement and the costs and benefits of seeking recovery, the Granted Shares. The Committee will determine in its discretion the amount, if any, the Company will seek to recover from the Participant. The Company may offset the recoupment amount against current or future incentive and non-incentive compensation and through cancellation of unvested or vested equity awards. In addition, the Committee may, to the extent permitted by law, take other remedial and recovery action, as determined by the Committee. The recoupment of incentive compensation under this provision is in addition to any other right or remedy available to the Company.

2.2 General Restrictions on Transfer of Granted Shares.

If in connection with a registration statement filed by the Company pursuant to the Securities Act of 1933, as amended (the “1933 Act”), the Company or its underwriter so requests, the Participant will agree not to sell any of his or her Granted Shares whether or not the Lapsing Forfeiture Right has lapsed for a period not to exceed the lesser of: (i) 210 days following the effectiveness of such registration statement or (ii) such period as the officers and directors of the Company agree not to sell their Common Stock of the Company.

(b) The Participant acknowledges and agrees that neither the Company, nor its shareholders nor its directors and officers, has any duty or obligation to disclose to the Participant any material information regarding the business of the Company or affecting the value of the Granted Shares before, at the time of, or following a Termination, including, without limitation, any information concerning plans for the Company to make a public offering of its securities or to be acquired by or merged with or into another firm or entity.

3. Securities Law Compliance. The Participant specifically acknowledges and agrees that any sales of Granted Shares shall be made in accordance with the requirements of the 1933 Act.

4. Rights as a Stockholder. The Participant shall have all the rights of a stockholder with respect to the Granted Shares, including voting and dividend rights, subject to the transfer and other restrictions set forth herein, including pursuant to Section 2.1(c) hereof, and in the Plan.

5. Legend. In addition to any legend required pursuant to the Plan, all certificates representing the Granted Shares to be issued to the Participant pursuant to this Agreement shall have endorsed thereon a legend substantially as follows:

“The shares represented by this certificate are subject to restrictions set forth in a Restricted Stock Agreement dated as of June 1, 2021 with this Company, a copy of which Agreement is available for inspection at the offices of the Company or will be made available upon request.”

6. Incorporation of the Plan. The Participant specifically understands and agrees that the Granted Shares issued under the Plan are being issued to the Participant pursuant to the Plan, a copy of which Plan the Participant acknowledges he or she has read and understands and by which Plan he or she agrees to be bound. The provisions of the Plan are incorporated herein by reference.

7. Tax Liability of the Participant and Payment of Taxes. The Participant acknowledges and agrees that any income or other taxes due from the Participant with respect to the Granted Shares issued pursuant to this Agreement, including, without limitation, the Lapsing Forfeiture Right, shall be the Participant’s responsibility. Without limiting the foregoing, the Participant agrees that, to the extent that the lapsing of restrictions on disposition of any of the Granted Shares or the declaration of dividends on any such shares before the lapse of such restrictions on disposition results in the Participant’s being deemed to be in receipt of earned income under the provisions of the Code, the Company shall be entitled to immediate payment from the Participant of the amount of any tax required to be withheld by the Company.

tion of this Agreement, the Participant may file an election under Section 83 of the Code. The Participant acknowledges that if he or she does not file such an election, as the Granted Shares are released from the Lapsing Forfeiture Right in accordance with Section 2.1, the Participant will have income for tax purposes equal to the fair market value of the Granted Shares at such date, less the price paid for the Granted Shares by the Participant.

8. Equitable Relief. The Participant specifically acknowledges and agrees that in the event of a breach or threatened breach of the provisions of this Agreement or the Plan, including the attempted transfer of the Granted Shares by the Participant in violation of this Agreement, monetary damages may not be adequate to compensate the Company, and, therefore, in the event of such a breach or threatened breach, in addition to any right to damages, the Company shall be entitled to equitable relief in any court having competent jurisdiction. Nothing herein shall be construed as prohibiting the Company from pursuing any other remedies available to it for any such breach or threatened breach.

9. No Obligation to Maintain Relationship. The Company is not by the Plan or this Agreement obligated to continue the Participant as an Employee, director or Consultant of the Company or an Affiliate of the Company. The Participant acknowledges: (i) that the Plan is discretionary in nature and may be suspended or terminated by the Company at any time; (ii) that the grant of the Granted Shares is a one-time benefit which does not create any contractual or other right to receive future grants of shares, or benefits in lieu of shares; (iii) that all determinations with respect to any such future grants, including, but not limited to, the times when shares shall be granted, the number of shares to be granted, the purchase price, and the time or times when each share shall be free from a lapsing forfeiture right, will be at the sole discretion of the Company; (iv) that the Participant’s participation in the Plan is voluntary; (v) that the value of the Granted Shares is an extraordinary item of compensation; and (vi) that the Granted Shares are not part of normal or expected compensation for purposes of calculating any severance, resignation, redundancy, end of service payments, bonuses, long-service awards, pension or retirement benefits or similar payments.

10. Notices. Any notices required or permitted by the terms of this Agreement or the Plan shall be given by recognized courier service, facsimile, registered or certified mail, return receipt requested, addressed as follows:

If to the Company:

Aspen Aerogels, Inc.
30 Forbes Road, Building B

Northborough, MA 01532
Attn: Stock Plan Administrator

If to the Participant:

C/o Aspen Aerogels, Inc.
30 Forbes Road, Building B
Northborough, MA 01532
Attn: Donald R. Young

or to such other address or addresses of which notice in the same manner has previously been given. Any such notice shall be deemed to have been given on the earliest of receipt, one business day following delivery by the sender to a recognized courier service, or three business days following mailing by registered or certified mail.

11. Benefit of Agreement. Subject to the provisions of the Plan and the other provisions hereof, this Agreement shall be for the benefit of and shall be binding upon the heirs, executors, administrators, successors and assigns of the parties hereto.

12. Governing Law. This Agreement shall be construed and enforced in accordance with the laws of the State of Delaware, without giving effect to the conflict of law principles thereof. For the purpose of litigating any dispute that arises under this Agreement, whether at law or in equity, the parties hereby consent to exclusive jurisdiction in Massachusetts and agree that such litigation shall be conducted in the courts of the Commonwealth of Massachusetts or the federal courts of the United States for the District of Massachusetts.

13. Severability. If any provision of this Agreement is held to be invalid or unenforceable by a court of competent jurisdiction, then such provision or provisions shall be modified to the extent necessary to make such provision valid and enforceable, and to the extent that this is impossible, then such provision shall be deemed to be excised from this Agreement, and the validity, legality and enforceability of the rest of this Agreement shall not be affected thereby.

14. Entire Agreement. This Agreement, together with the Plan, constitutes the entire agreement and understanding between the parties hereto with respect to the subject matter hereof and supersedes all prior oral or written agreements and understandings relating to the subject matter hereof. No statement, representation, warranty, covenant or agreement not expressly set forth in this Agreement shall affect or be used to interpret, change or restrict the express terms and provisions of this Agreement provided, however, in any event, this Agreement shall be subject to and governed by the Plan.

15. Modifications and Amendments; Waivers and Consents. The terms and provisions of this Agreement may be modified or amended as provided in the Plan. Except as provided in the Plan, the terms and provisions of this Agreement may be waived, or consent for the departure therefrom granted, only by written document executed by the party entitled to the benefits of such terms or provisions. No such waiver or consent shall be deemed to be or shall constitute a waiver or consent with respect to any other terms or provisions of this Agreement, whether or not similar. Each such waiver or consent shall be effective only in the specific instance and for the purpose for which it was given, and shall not constitute a continuing waiver or consent.

16. Consent of Spouse/Domestic Partner. If the Participant has a spouse or domestic partner as of the date of this Agreement, the Participant's spouse or domestic partner shall execute a Consent of Spouse/Domestic Partner in the form of Exhibit A hereto, effective as of the date hereof. Such consent shall not be deemed to confer or convey to the spouse or domestic partner any rights in the Granted Shares that do not otherwise exist by operation of law or the agreement of the parties. If the Participant subsequent to the date hereof, marries, remarries or applies to the Company for domestic partner benefits, the Participant shall, not later than 60 days thereafter, obtain his or her new spouse/domestic partner's acknowledgement of and consent to the existence and binding effect of all restrictions contained in this Agreement by having such spouse/domestic partner execute and deliver a Consent of Spouse/Domestic Partner in the form of Exhibit A.

17. Counterparts. This Agreement may be executed in one or more counterparts, and by different parties hereto on separate counterparts, each of which shall be deemed an original, but all of which together shall constitute one and the same instrument.

18. Data Privacy. By entering into this Agreement, the Participant: (i) authorizes the Company and each Affiliate, and any agent of the Company or any Affiliate administering the Plan or providing Plan record keeping services, to disclose to the Company or any of its Affiliates such information and data as the Company or any such Affiliate shall request in order to facilitate the grant of Shares and the administration of the Plan; and (ii) authorizes the Company and each Affiliate to store and transmit such information in electronic form for the purposes set forth in this Agreement.

[THE NEXT PAGE IS THE SIGNATURE PAGE]

IN WITNESS WHEREOF, the parties hereto have executed this Agreement as of the day and year first above written.

ASPEN AEROGELS, INC.

By: /s/ John F. Fairbanks
Name: John F. Fairbanks
Title: VP, CFO and Treasurer

Participant:

/s/ Donald R. Young
Donald R. Young

CONSENT OF SPOUSE/DOMESTIC PARTNER

I, _____, spouse or domestic partner of _____, acknowledge that I have read the RESTRICTED STOCK AGREEMENT dated as of _____, 20__ (the "Agreement") to which this Consent is attached as Exhibit A and that I know its contents. Capitalized terms used and not defined herein shall have the meanings assigned to such terms in the Agreement. I am aware that by its provisions the Granted Shares granted to my spouse/domestic partner pursuant to the Agreement are subject to a Lapsing Forfeiture Right in favor of Aspen Aerogels, Inc. (the "Company") and that, accordingly, I may be required to forfeit to the Company any or all of the Granted Shares of which I may become possessed as a result of a gift from my spouse/domestic partner or a court decree and/or any property settlement in any domestic litigation.

I hereby agree that my interest, if any, in the Granted Shares subject to the Agreement shall be irrevocably bound by the Agreement and further understand and agree that any community property interest I may have in the Granted Shares shall be similarly bound by the Agreement.

I agree to the Lapsing Forfeiture Right described in the Agreement and I hereby consent to the forfeiture of the Granted Shares to the Company by my spouse/domestic partner or my spouse/domestic partner's legal representative in accordance with the provisions of the Agreement. Further, as part of the consideration for the Agreement, I agree that at my death, if I have not disposed of any interest of mine in the Granted Shares by an outright bequest of the Granted Shares to my spouse or domestic partner, then the Company shall have the same rights against my legal representative to exercise its rights to the Granted Shares with respect to any interest of mine in the Granted Shares as it would have had pursuant to the Agreement if I had acquired the Granted Shares pursuant to a court decree in domestic litigation.

I AM AWARE THAT THE LEGAL, FINANCIAL AND RELATED MATTERS CONTAINED IN THE AGREEMENT ARE COMPLEX AND THAT I AM FREE TO SEEK INDEPENDENT PROFESSIONAL GUIDANCE OR COUNSEL WITH RESPECT TO THIS CONSENT. I HAVE EITHER SOUGHT SUCH GUIDANCE OR COUNSEL OR DETERMINED AFTER REVIEWING THE AGREEMENT CAREFULLY THAT I WILL WAIVE SUCH RIGHT.

Dated as of the _____ day of _____, 20__.

Print name:

THIRD AMENDMENT TO EXECUTIVE AGREEMENT

This Amendment (the “Amendment”), effective as of June 29, 2021, is entered into by and between Aspen Aerogels, Inc., a Delaware corporation (the “Company”), and Donald R. Young (the “Executive”), for purposes of amending that certain Executive Agreement between the parties effective as of January 1, 2019 as previously amended on August 1, 2019 and on January 1, 2020 (the “Executive Agreement”).

RECITALS

Whereas, the Company has determined to grant the Executive an equity award (the “Award”) pursuant to that certain Performance-Based Restricted Stock Agreement, by and between the Company and the Executive as of the date hereof (the “Restricted Stock Agreement”); and

Whereas, in connection with the Award granted under the Restricted Stock Agreement, the Company and the Executive wish to amend the Executive Agreement to provide for the removal of the bonus payable to the Executive in the event a Change of Control is consummated within the Performance Period and the CIC Price is equal to or greater than \$6.00; and

Now, therefore, in consideration of mutual promises and other terms set forth below, the Company and the Executive agree as follows:

1. Amendment. The Executive Agreement is hereby amended as follows:
 - (A) The text of Section 2(d)(i) is deleted in full and shall now provide that it is “RESERVED.”
 - (B) The text of Section 2(d)(ii) is deleted in full and shall now provide that it is “RESERVED.”
 - (C) The text of Section 2(d)(iii) is deleted in full and shall now provide that it is “RESERVED.”
 - (D) The text of Section 2(d)(v) is deleted in full and shall now provide that it is “RESERVED.”

 2. Miscellaneous Provisions. Capitalized terms used but not defined within this Amendment have the meanings specified within the Executive Agreement. Except as expressly modified by this Amendment, all terms and conditions of the Executive Agreement remain in full force and effect. This Amendment can be executed by the parties hereto by facsimile and in counterparts, each of which is deemed an original and both of which taken together constitutes one agreement binding upon the parties hereto. This Amendment shall be governed by and construed in accordance with the laws of the Commonwealth of Massachusetts, without giving effect to any choice of law or conflict of law provision or rule (whether of the Commonwealth of Massachusetts or any other jurisdiction) that would cause the application of the law of any jurisdiction other than the Commonwealth of Massachusetts.
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In witness whereof, the Parties hereto enter into this Amendment by signing below.

ASPEN AEROGELS, INC.

By: /s/ John F. Fairbanks

Name: John F. Fairbanks

Title: Vice President, Treasurer, and Chief Financial Officer

THE EXECUTIVE

By: /s/ Donald R. Young

Name: Donald R. Young

June 29, 2021

Date

CERTIFICATIONS UNDER SECTION 302

I, Donald R. Young, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: August 4, 2021

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

CERTIFICATIONS UNDER SECTION 302

I, John F. Fairbanks, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Aspen Aerogels, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: August 4, 2021

/s/ John F. Fairbanks

John F. Fairbanks

Vice President, Chief Financial Officer and Treasurer (principal financial officer and principal accounting officer)

CERTIFICATIONS UNDER SECTION 906

Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002 (subsections (a) and (b) of section 1350, chapter 63 of title 18, United States Code), each of the undersigned officers of Aspen Aerogels, Inc., a Delaware corporation (the “Company”), does hereby certify, to such officer’s knowledge, that:

The Quarterly Report on Form 10-Q for the quarter ended June 30, 2021 (the “Form 10-Q”) of the Company fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934, and the information contained in the Form 10-Q fairly presents, in all material respects, the financial condition and results of operations of the Company.

Dated: August 4, 2021

/s/ Donald R. Young

Donald R. Young
President and Chief Executive Officer
(principal executive officer)

Dated: August 4, 2021

/s/ John F. Fairbanks

John F. Fairbanks
Vice President, Chief Financial Officer and Treasurer
(principal financial officer and principal accounting officer)

A signed original of this written statement required by Section 906 has been provided to the Company and will be retained by the Company and furnished to the Securities and Exchange Commission or its staff upon request.